DBC 1200

Pressure-Build Electro-Hydraulic Control Unit

ECU Diagnostic Communication

for ISO 14229

Customer Technical Document

Specification # 18116030

Rev 1.2

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Revision History

Doc Version	Date	Change Summary
0.1	12/14/2011	Initial Draft for DBC1200
0.2	3/22/2011	Outlined DIDs for specific 12v content; Added supplier reserved DIDs.
0.3	5/3/2012	Updated the DID Table 4.3-1 With New/Revised DIDs and R/W Information
0.4	11/12/12	Updated DID F18C and 3001 with details in the table
0.5	11/26/12	Updated DID Table and Provided detail for 3002, 3004, 3005, 3006, 301C, 3034 and Revised section 2.2.1 and 2.2.2.
0.6	11/28/12	Revised Device Control section 4.3.3.1 and 4.3.3.2
1.0	2/13/13	Section 4: Format and content updates to tables associated with DIDs $3008, $300A, $300B, $300C, $3001, $F18C, $3002, $3004, $3005, $3006, $3034, $3011 Added $301C to supplier reserved DIDs Format changes to tables in Input/Output Control sections 4.3.3.2 and 4.3.3.1 Updated list of Supplier Reserved DIDs. Updated list of Diagnostic Services in 2.3.1 Added part number and removed DRAFT watermark.
1.1	3/1/13	Revised Sid $22 positive and negative responses.
1.2	4/25/13	Revised table 111 by adding byte 4,5 to ECU Bit fault table DID $3009

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1 Introduction

This document is intended as a reference document that describes diagnostic functionality and diagnostic communication services of the DBC 1200 ECU. This document is NOT the specification for the diagnostic functionality of the DBC 1200 ECU.

1.1 Scope of Document

This document describes the technical details of on-board assembly line and service diagnostics for DBC 1200 applications that utilize ISO 14229 requirements. It is to be used in conjunction with following documentation provided for the DBC1200 PB EHCU.

DBC 1200 - Pressure-Build Electro-Hydraulic Control Unit, Use and Application Customer Technical Document

DBC 1200 - Pressure-Build Electro-Hydraulic Control Unit, Diagnostic and Service Manual

DBC1200 Customer Diagnostic Code Cross Reference Table (Note: This document is unique for each customer application)

DBC1200 Customer Mechanization Drawing (Note: This document is unique for each customer application)

1.2 Mission/Theme

The DBC 1200 ECU is equipped with integrated self diagnostic functionality capable of determining failures in the EHCU and its external inputs (i.e. sensors, etc.). The ECU uses the self diagnostic functionality to prevent damage to components and to ensure that the ECU behaves in a predictable and safe manner in the presence of malfunctions. The ECU is equipped with additional capabilities to support ECU and EHCU manufacturing, vehicle assembly activities as well as testing in a vehicle assembly or service environment.

1.3 Order of Precedence

In the event of a conflict between the text of this document and the documents cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

This document is NOT the controlling specification for the features and functionality described within.

1.4 Applicable Documents

See section 11 of "DBC 1200 - Pressure-Modulation Electro-Hydraulic Control Unit, Use and Application Customer Technical Document", Specification number, 18111832.

1.5 References

See section 12 of "DBC 1200 - Pressure-Build Electro-Hydraulic Control Unit, Use and Application Customer Technical Document", Specification number, 18111832.

2 DBC 1200 ECU Functionality

2.1 Diagnostic Protocol

2.1.1 Physical Layer

The High Speed CAN bus physical layer conforms to the requirements defined in ISO 11898-2:Road vehicles — Controller area network (CAN) — Part 2: High-speed medium access unit. Refer to that document for details.

2.1.2 Data Link Layer

The data link layer is implemented according to ISO 11898-1: Road vehicles — Controller area network (CAN) — Part 1: Data link layer and physical signaling.

2.1.2.1 Diagnostic Message CAN Identifiers

Diagnostic CAN identifiers shall be implemented according to the vehicle communication requirements for the customer vehicle application.

2.1.3 Network/transport layer

The network layer is implemented according to ISO 15765-2:Road vehicles — Diagnostics on controller area network (CAN) — Part 2: Network layer service.

2.2 Self Test Operation

Intialization self test and Continuos self tests have been developed to detect abnormal part operation. These tests include Initialization self tests and Continous self test routines. These test routines are outlined in sections 2.2.1 and 2.2.2 below. Timing requirements for the individual tests have been identified in those sections as well. The total test time to complete the routines is estimated below with more detail following in the sections below.

Total Test Time Estimate:

Based on assembly line requirements, the total test time for the after ignition initialization and a complete run of all the continuous self-tests shall not take longer than 2.5 seconds. Does not include conditional actuator test time.

Additional test coverage can be gained by ensuring that the conditions are met to run the actuator control tests and allowing time for the actuator control tests to run.

2.2.1 Initialization Self Test

There are two main groupings of initialization self tests. Those tests that run immediately after the ignition is turned on and additionally, the actuator control tests which are run when specific external conditions exist as identified below:

• Tests After Ignition is Enabled:

- The initialization self-test routines run immediately after the ignition is turned on and will be performed in less than 500 ms.

• Conditional Actuator Control Tests:

- Before the actuator control tests are initiated, the following external conditions are required:

- ((Engine rpm >= 500 rpm OR Vehicle Speed >=5kph) & Brake=OFF)) OR ((Vehicle Speed >= 16 kph) & (Brake=ON))

- Once initiated, the actuator control tests will complete in less than 3000 ms.

2.2.2 Continuous Self Test Operation

All continuous self-test routines must be performed at least once during a total time period, which is less than 2 seconds. The self-test routines are performed simultaneously.

2.2.3 Fault Definition

A fault is a malfunction, which will cause noticeable abnormal system behavior, performance problems or damage. Additional system faults which may not have a noticeable, adverse effect can be detected in support of customer safety or compliance to governmental safety standards.

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2.2.3.1 DTC Number

The numbering of Diagnostic Trouble Codes (DTC) will be made per the C0XXX standard for the DBC1200 ECU. Refer to the Customer Application Diagnostic Code Cross Reference. This document will list the following:

1. The applicable diagnostic tests for a customer application vehicle.
2. The corresponding DBC1200 Fault Code Number in C0XXX format
3. Any customer specific fault code reporting formats that are used for diagnostic communications.
4. Any special latching / recovery limitations for diagnostics

2.2.3.2 DTC Description

The descriptions of the DTC are defined in the document "DBC1200 Pressure Build EHCU Diagnostic and Service Manual". The document includes detailed descriptions of:

1. Failure type definitions
2. Enable criteria including vehicle power modes, ECU operating modes
3. Secondary fault setting criteria and exceptions.
4. Fault Actions.

2.2.4 General Diagnostic Fault Processing

The following paragraphs outline the basic process for all diagnostic trouble code processing.

2.2.4.1.1 Fault Detection/Test

A diagnostic test is defined as: Actions performed from a given moment when all conditions to start a diagnostic fault detection algorithm are fulfilled until the moment a diagnostic fault detection algorithm makes a pass or fail decision.

Diagostic tests should report faults only when a real fault condition exists. In order to prevent false failure conditions, all necessary logic (e.g. necessary filtering) to determine whether a real fault exists will be performed before a result is determined.

Secondary failures are generally not logged. Wherever possible the diagnostic fault detection algorithms will be designed such that a single point failure results in a single failure being logged. Exceptions will be documented in the "DBC1200 Pressure-Build Electro-Hydraulic Control Unit Diagnostics and Service Manual".

When a diagnostic trouble code (DTC) has been asserted due to a failure detected in the current ignition cycle, the DTC will be stored to NVM (Non-Volatile Memory). The DTC will then be reported with a status of "history" on subsequent ignition cycles.

The figure below illustrates the processing of a DTC that occurs during an ignition cycle:

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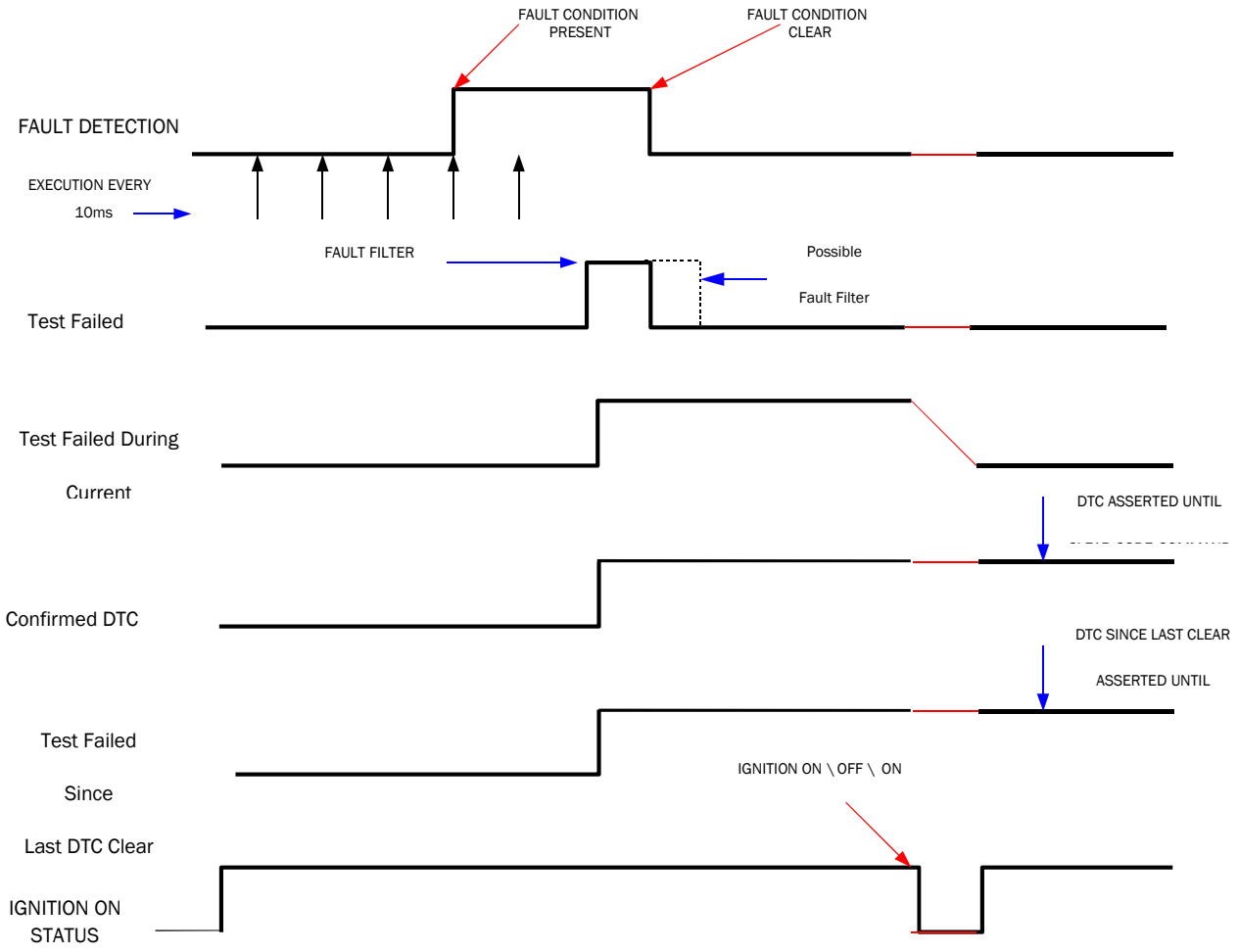

Figure 1 - DBC 1200 DTC Processing

2.2.4.1.2 Indication to Vehicle Operator

If information concerning a fault is judged to be important for the vehicle operator, the fault information will typically be indicated by illuminating a warning lamp. Specific fault indication actions are described in the “DBC1200 Diagnostic and Service Manual”. Fault indiciatiions are usually transmitted by serial daat from the DBC 1200 ECU to another vehicle ECU responsible for controlling lamps / telltales and other driver information.

The vehicle operator indication will remain present for as long as the detected fault condition is present. DTCs that do not have an active fault condition, but are stored with a status of “history” will not generate a fault indication to the vehicle operator.

2.2.4.1.3 Clearing DTC History

The deletion of a DTC with a status of “history” will be possible with the diagnostic service ”ClearDiagnosticInformation”.

In addition the DBC1200 EHCU will implement an ”Automatic Fault Recovery” function.

In general, the Automatic Fault Recovery function will operate as follows:

The fault information for a specific diagnostic test will be cleared after 100 consecutive drive cycles with no assertions of any other Fault Test conditions (Note: the drive cycle threshold value can be customized based on customer requirements). If a Failure Record has been stored, the content of the failure record will be cleared and made available for a new failure record storage by any Fault Test. A drive cycle occurs after vehicle speed exceeds a minium threshold (typical value is 16 kph) during the power ON period, and the ECU transitions from the ON to OFF state

The figure below illustrates the Automatic Fault Recovery function:

2.2.4.1.4 DTC Information Storage

The diagnostic trouble codes will be stored in non-volatile memory (NVM). When a SId $19 (ReportDTCInformation) request is received by the ECU, the DTC number and associated information will then be reported to the diagnostic test tool.

Note that battery interuptions that occur after the DTC has been initialiy reported, may prevent the fault information for that ignition cycle from being stored to NVM.

2.2.4.1.5 Failure Records and Enhanced Diagnostics

The DBC1200 PB EHCU has the capability to store three enhanced failure records in non volatile memory (EEPROM).

The failure records will include various system and vehicle conditions at the time when a fault code occurred, when last fault last occurred, the second last fault last occurred, and so on.

This feature is known as enhanced diagnostics. The content of the enhanced diagnostic failure records is described in the Sections 4.3.2.12, 4.3.2.13, and 4.3.2.14

When a DTC that does not have an existing failure record is recorded, the failure record information at the time of DTC setting will be captured. If the DTC already has a failure record from a previous ignition cycle, then only the “Ignition Cycles Since Last Fault Code Set” and “Number of Occurrences of DTC” will be updated. All other data will reflect the data values from the first time the code was set.

The last three DTCs which support failure records that occur in an ignition cycle will have failure records stored. If additional DTCs are asserted during the ignition cycle, only the 3 most recent DTCs will have failure records stored. A new failure record will be placed into an empty failure record, if available. If all failure record areas are populated with data from previous ignition cycles when a new failure is recorded, the new DTC and its failure record information will replace the oldest failure record’s information.

Executing a clear codes diagnostic message operation will clear the failure record information for any DTC event, including all its stored failure record data. The newly cleared failure record area will be made available for data storage by any new DTC.

2.3 Diagnostic Services

Diagnostic Services will be implemented mainly according to ISO 14229 – 1; Additional details are described in this section.

2.3.1 DBC 1200 Supported Service Identifier value summary table

The purpose of the "DBC 1200 Supported Service Identifier value summary table" is to provide an overview about all services specified/referenced in this document. The available services depend on if the ECU is operating in the normal application software mode, or if the ECU is only operating with boot software. Any other attempted service request would be responded with serviceNotSupported (SNS, $11).

Diagnostic Services	SId	Run Time Diagnostic Mode	Boot / Programming Mode
DiagnosticSessionControl	10	X	X
ClearDiagnosticInformation	14	X	
ECUResetRequest	11		X
ReadDTCInformation	19	X	
ReadDataByIdentifier	22	X	X
ReadMemoryByAddress	23	X	X
Security Access	27	X	X
CommunicationControl	28	X	X
WriteDataByIdentifier	2E	X	
InputOutputControlByIdentifier	2F	X	
RequestDownload	34	X	X
Transfer Data	36	X	X
Request Transfer Exit	37		X
TesterPresent	3E	X	X
TestService	B0	X	

Table 2 - DBC 1200 Supported Service Identification Value Summary Table

2.3.2 Diagnostic and Communication Management functional unit

In this unit DBC 1200 supports the diagnostic services listed in the following table,

Service	Description
DiagnosticSessionControl	The client requests to control a diagnostic session with a server(s).
SecurityAccess	The client requests to unlock a secured server(s).
TesterPresent	The client indicates to the server(s) that it is still present.
CommunicationsControl	The client requests the server to control its communication.

Table 3 - Diagnostic and Diagnostic Management functional unit

2.3.2.1 DiagnosticSessionControl Service (SId $10)

This service is used to enable different diagnostic Sessions in the server. DBC 1200 uses this service to enter safetySystemDiagnosticSession (DVT mode), ExtendedDiagnsoticSession or go back to the defaultSession.

When the system is in the safetySystemDiagnosticSession.

1. Most of the system control functions like ABS are disabled.

2. DRP will remain enabled unless the I/O Control Service InputOutputControlByIdentifier (Sid $2F)is active

3. The appropriate system status signals / commands indicating the DRP, ABS functional statuses will be sent

4. Normal diagnostic (DTC) algorithms will be run until the I/O Control Service InputOutputControlByIdentifier (Sid $2F) is active. This will guarantee that all DTC data is accurate when the test tool checks for DTC's prior to the first I/O Control Service.

5. InputOutputControlByIdentifier ($2F) may be entered at any vehicle speed.

2.3.2.1.1 SId $10 - Request Message definition

The elements of the request message are described in the following paragraphs.

2.3.2.1.1.1 SId $10 - Sub-function Parameter Level Definition

The sub-function parameter diagnosticSessionType is used by the DiagnosticSessionControl service to select the specific behaviour of the server. Explanations and usage of the possible diagnostic sessions are detailed below. The following sub-function values are specified (suppressPosRspMsgIndicationBit (bit 7) not shown)

Diagnostic session transition description: When the server transitions from diagnostic session to diagnostic session including the currently active diagnostic session, the server shall re-initialize completely. The server shall reset all activated/initiated/changed settings/controls during the activated session. This does not include long term changes programmed into non-volatile memory.

In all but the default diagnostic session the tester present timer is activated Any request of services listed in Table 3 - DBC 1200 Supported Service Identification Value Summary Table shall be responded with a negative code $22 if it is not available under currently active diagnostic session.

Hex	Description	Mnemonic
01	defaultSession This diagnostic session enables the default diagnostic session in the server(s) and does not support any diagnostic application timeout handling provisions	DS
02	programmingSeession	PRGM
03	extendedDiagnosticSession This diagnosticSession can e.g. be used to enable all diagnostic services required to support the adjustment of functions. It can also be used to enable diagnostic services, which are not specifically tied to the adjustment of functions.	EXTDS
04	safetySystemDiagnosticSession This diagnosticSession enables all diagnostic services required to support safety system related functions e.g. airbag deployment.	SSDS

Table 4 – SId $10 Definition diagnosticSession Values

2.3.2.1.1.1.1 DefaultSession

If no diagnostic session has been requested by the client after system initialization default session shall be automatically enabled in the server. The default session shall support the following services.

Diagnostic Services	Sid
DiagnosticSessionControl	10
ClearDiagnosticInformation	14
ReadDTCInformation	19
ReadDataByIdentifier	22
ReadMemoryByAddress	23
SecurityAccess	27
WriteDataByIdentifier	2E
TesterPresent	3E
Test	B0

Table 5 - DBC 1200 Supported Default Service Identification Value Summary Table

2.3.2.1.1.1.2 Programming Session

In the programming session, the following diagnostic services shall be available.

Diagnostic Services	SId
DiagnosticSessionControl	10
ReadDataByIdentifier	22
ReadMemoryByAddress	23
SecurityAccess	27
WriteDataByIdentifier	2E
TesterPresent	3E
Test	B0

Table 6 - DBC 1200 ProgrammingSession Service Identification Value Summary Table

2.3.2.1.1.1.3 ExtendedDiagnosticSession

In the extended diagnostic session, the following diagnostic services shall be available.

Diagnostic Services	SId
DiagnosticSessionControl	10
ClearDiagnosticInformation	14
ReadDTCInformation	19
ReadDataByIdentifier	22
ReadMemoryByAddress	23
SecurityAccess	27
CommunicationsControl	28
WriteDataByIdentifier	2E
InputOutputControlByIdentifier	2F
TesterPresent	3E
Test	B0

Table 7 - ExtendedDiagnsoticSession Service Identification Value Summary Table

2.3.2.1.1.1.4 SafetySystemDiagnosticSession

Upon this diagnostic session active, the DBC 1200 shall be in the DVT (Dynamic Vehicle Test) mode.

When the system is in the safetySystemDiagnosticSession.

1. Most of the system control functions like ABS are disabled.

2. DRP will remain enabled unless the I/O Control Service InputOutputControlByIdentifier (Sid $2F) is active

3. The appropriate system status signals / commands indicating the DRP, ABS functional status will be sent

4. Normal diagnostic (DTC) algorithms will be run until the I/O Control Service InputOutputControlByIdentifier (Sid $2F) is active. This will guarantee that all DTC data is accurate when the test tool checks for DTC's prior to the first I/O Control Service.

5. InputOutputControlByIdentifier ($2F) may be entered at any vehicle speed.

The following diagnostic services shall be enabled.

Diagnostic Services	SId
DiagnosticSessionControl	10
ReadDTCInformation	19
ReadDataByIdentifier	22
ReadMemoryByAddress	23
InputOutputControlByIdentifier	2F
TesterPresent	3E

Table 8 - SafetySystemDiagnosticSession Service Identification Value Summary Table

2.3.2.1.1.2 SId $10 - Data parameter definition

This service does not support data parameters in the request message.

2.3.2.1.2 SId $10 - Request Message Data Bytes

See table below.

Data Byte	Parameter Name	Cvt	Hex Value	Mnemonic
#1	DiagnosticSessionContro Request Service Id	M	10	DSC
#2	sub-function = [defaultSession programmingSession extendedDiagnosticSession safetySystemDiagnosticSession]	M	 01 02 03 04	LEV- DS PRGM EXTDS-SSDS

Table 9 – SId $10 Request Message

2.3.2.1.3 SId $10 - Positive Response Message Data Bytes

See table below.

Data Byte	Parameter Name	Cvt	Hex Value	Mnemonic
#1	DDiagnosticSessionContro Positive Response Service Id	S	50	STDSPR
#2	DiagnosticSession= [defaultSession programmingSession extendedDiagnosticSession safetySystemDiagnosticSession]	M	 01 02 03 04	DCM DS PRGM EXTDS SSDS

Table 10 – SId $10 Positive Response Message

2.3.2.1.4 SId $10 - Negative Response Message Data Bytes

See tables below.

Data Byte	Parameter Name	Cvt	Hex Value	Mnemonic
#1	negativeResponse Service Id	S	7F	NR
#2	DiagnosticSessionContro Request Service Id	S	10	STDS
#3	ResponseCode = [subFunctionNotSupported incorrectMessageLengthOrInvalidFormat ConditionsNotCorrect]	M	 12 13 22	 SFNS IMLOIF CNC

Table 11 –Sid $10 Negative Response Message

Response Code (Hex)	Conditions
12	subFunctionNotSupported Requested DiagnosticSession is not supported
13	incorrectMessageLengthOrInvalidFormat Wrong length for the specific request
22	ConditionsNotCorrect This response code will occur if the node (ECU) internal conditions do not allow the change of diagnostic Session. The system is currently in initialization process. The system is in the process of changing EEPROM. The system has fault which causing the system relay to be turning off.

Table 12 – Sid $10 Response Codes

2.3.2.2 SecurityAccess Service (SId $27)

The purpose of this service is to provide a means to access data and/or diagnostic services, which have restricted access for security, emissions, or safety reasons.

A typical example of the use of this service is as follows:

- client requests the “Seed”
- server sends the “Seed”
- client sends the “Key” (appropriate for the Seed received)
- server responds that the “Key” was valid and that it will unlock itself

A vehicle manufacturer specific time delay (10 seconds) shall be required before the server can positively respond to a service SecurityAccess ‘requestSeed’ message from the client after server power up/reset and after a certain number of false access attempts (see further description below). The delay shall be activated after a vehicle manufacturer specified number of false access attempts has been reached or when the server is powered up/reset and a previously performed SecurityAccess service has failed due to a single false access attempt. After a successful SecurityAccess service 'sendKey' execution the server internal indication information for a delay timer invocation (we only support one pare of security seed and key for DID write protection at this time) on a power up/reset shall be set to FALSE by the server. In case the server cannot determine if a previously performed SecurityAccess service prior to the power up/reset has failed then the delay timer shall always be active after power up/reset. The delay is only required if the server is locked when powered up/reset.

The client shall request the server to "unlock" by sending the service SecurityAccess ‘requestSeed’ message. The server shall respond by sending a "seed" using the service SecurityAccess ‘requestSeed’ positive response message. The client shall then respond by returning a "key" number back to the server using the appropriate service SecurityAccess ‘sendKey’ request message. The server shall compare this "key" to one internally stored. If the two numbers match, then the server shall enable

("unlock") the client's access to specific services/data and indicate that with the service SecurityAccess ‘sendKey’ positive response message. If the two numbers do not match, this shall be considered a false access attempt. If access is rejected for any other reason, it shall not be considered a false access attempt. An invalid key requires the client to start over from the beginning with a SecurityAccess 'requestSeed' message.

If a server is already unlocked when a SecurityAccess ‘requestSeed’ message is received, that server shall respond with a SecurityAccess ‘requestSeed’ positive response message service with a seed value equal to zero (0). The client shall use this method to determine if a server is locked by checking for a non-zero seed.

There shall always be a fixed relationship for each level of security supported so that the sendKey sub-function parameter value used for any given security level shall be equal to the requestSeed sub-function parameter value used for that security level plus one.

Only one security level shall be active at any instant of time. For example, if the security level associated with requestSeed 03 hex is active and a tester request is successful in unlocking the security level associated with requestSeed 01 hex, then only the secured functionality supported by the security level associated with requestSeed 01 hex shall be unlocked at that time. Any additional secured functionality that was previously unlocked by the security level associated with requestSeed 03 hex shall no longer be active. The security levels numbering is arbitrary and does not imply any relationship between the levels.

Attempts to access security shall not prevent normal vehicle communications or other diagnostic communication.

Servers, which provide security shall support reject messages if a secure service is requested while the server is locked.

Some diagnostic functions/services requested during a specific diagnostic session may require a successful security access sequence. In such case the following sequence of services shall be required:

- DiagnosticSessionControl service
- SecurityAccess service
- Secured diagnostic service

There are different accessModes allowed for an enabled diagnosticSession (session started) in the server.

Please note that some customer applications will only use one set of seed and key pair for both DID write and reprogramming securities. Others will use each set for one single purpose.

This service is only supported if requested by in Physical addressing mode.

2.3.2.2.1 SId $27 - Request Message Definition

The elements of the request message are described in the following paragraphs.

2.3.2.2.1.1 SId $27 - Sub-function Parameter Level Definition

The sub-function parameter securityAccessType indicates to the server the step in the process for this service, the level of security the client wants to access and the format of the seed and key.

Examples:

- “requestSeed=01 hex” identifies a fixed relationship between “requestSeed=01 hex” and “sendKey=02 hex”
- “requestSeed=03 hex” identifies a fixed relationship between “requestSeed=03 hex” and “sendKey=04 hex”

Values are defined in 2.3.1-1 for requestSeed and sendKey (suppressPosRspMsgIndicationBit (bit 7) not shown).

Hex (bit 6-0)	Description	Cvt	Mnemonic
61	requestSeed Request DID write protection Seed	U	RSD
62	Send Key Send DID write protection Key	U	SK
01	requestSeed Request programming Seed Request DID write protection Seed	U	RSD
02	Send Key Send Programming Key Send DID write protection Key	U	SK

Table 13 - SId $27 Request message sub-function parameter definition

2.3.2.2.1.2 SId $27 - Data parameter Definition

This service does not support data parameters in the request message.

2.3.2.2.2 SId $27 - Request Message Data Bytes

See table below.

A_PDU parameter	Parameter Name	Cvt	Hex Value	Mnemonic
#1	SecurityAcces Request Service Id	M	27	SA
#2	sub-function = [securityAccessType = requestSeed]	M	01,61	LEV_ SAT_RSD

Table 14 – SId $27 Request message definition - sub-function = requestSeed

A_PDU parameter	Parameter Name	Cvt	Hex Value	Mnemonic
#1	SecurityAcces Request Service Id	M	27	SA
#2	sub-function = [securityAccessType = sendKey]	M	02, 62	LEV_ SAT_SK
#3 #4	securityKey[] = [key#1 (high byte) key#2 (low byte)]	M M	00-FF 00-FF	SECKEY_ KEYHB KEYLB

Table 15 – SId $27 Request message definition - sub-function = sendKey

2.3.2.2.3 SId $27 - Positive Response Message

See table below.

A_PDU parameter	Parameter Name	Cvt	Hex Value	Mnemonic
#1	SecurityAccess Response Service Id	S	67	SAPR
#2	securityAccessType	M	01, 61	SAT_
	securitySeed[] = [			SECSEED_
#3	seed#1 (highbyte)	C	00-FF	SEED1HB
#4	seed#m (low byte)]	C	00-FF	SEED2LB

Table 16 - SId $27 requestSeed Positive Response Message

A_PDU parameter	Parameter Name	Cvt	Hex Value	Mnemonic
#1	SecurityAccess Response Service Id	S	67	SAPR
#2	securityAccessType	M	02,62	SAT_

Table 17 - SId $27 sendKey Positive Response Message

2.3.2.2.4 SId $27 - Negative Response Message

See tables below.

A_PDU parameter	Parameter Name	Cvt	Hex Value	Mnemonic
#1	negativeResponse Service Id	S	7F	NR
#2	testerPresent Request Service Id	M	27	TP
#3	ResponseCode = [	M		RC
	subFunctionNotSupported		12	SFNS
	incorrectMessageLengthOrInvalidFormat		13	IMLOIF
	ConditionsNotCorrect		22	CNC
	requestSequenceError		24	RSE
	invalidKey		35	IK
	exceededNumberOfAttempts		36	ENOA
	requiredTimeDelayNotExpired		37	RTDNE
	serviceNotSupportedInActiveSession]		7F	SNSIAS

Table 18 - SId $27 Negative Response Message

Response Code (hex)	Condition
12	subFunctionNotSupported Requested sub-function is not supported
13	incorrectMessageLengthOrInvalidFormat Wrong length for the specific request (Note: BWI can support configuring software to check for only length shorter than specified)
22	ConditionsNotCorrect Request is functional addressed
24	requestSequenceError Send if the 'sendKey' sub-function is received without first receiving a 'requestSeed' request message.
35	invalidKey Send if an expected 'sendKey' sub-function value is received and the value of the key does not match the server's internally stored/calculated key.
36	exceededNumberOfAttempts Send if the delay timer is active due to exceeding the maximum number of allowed false access attempts.
37	requiredTimeDelayNotExpired Send if the delay timer is active and a request is transmitted.
7F	serviceNotSupportedInActiveSession The service is not enabled in the current diagnostic session.

Table 19 - SId $27 Response Code

2.3.2.3 TesterPresent Service (SId $3E)

This service shall be used to indicate to a server that the client is present. This service is required in the absence of other service requests to prevent servers from automatically returning to default diagnostic session.. With each received TesterPresent service or any other services that send to DBC 1200, the DBC 1200 shall reset its S3 server timer. The presence of this service shall indicate that the system should remain in a diagnostic Session of operation.

2.3.2.3.1 SId $3E - Request Message definition

There are no parameters required in this diagnostic service.

2.3.2.3.1.1 SId $3E - Sub-function Parameter Level Definition

Support 00/80 sub-function. Byte value "00" can receive positive response and value "80" suppress Positive Response.

2.3.2.3.1.2 SId $3E - Data Parameter Definition

This service does not support data parameters in the request message.

2.3.2.3.2 SId $3E - Request Message Data Bytes

See table below.

A_PDU parameter	Parameter Name	Cvt	Hex Value	Mnemonic
#1	testerPresent Request Service Id	M	3E	TP
#2	Response Required = [zeroSubFunction]	M	00/80	LEV_ ZSUBF

Table 20 – SId $3E Request Message

2.3.2.3.3 SId $3E - Positive Response Message Data Bytes

See table below.

Data Byte	Parameter Name	Cvt	Hex Value	Mnemonic
#1	testerPresent Positive Response Service Id	S	7E	TPPR
#2	Response Required = [zeroSubFunction]	M	00	ZSUBF

Table 21 - SId $3E Positive Response Message

2.3.2.3.4 SId $3E - Negative Response Message Data Bytes

See tables below.

Data Byte	Parameter Name	Cvt	Hex Value	Mnemonic
#1	negativeResponse Service Id	S	7F	NR
#2	testerPresent Request Service Id	M	3E	TP
#3	ResponseCode = [subFunctionNotSupported incorrectMessageLengthOrInvalidFormat]	M	 12 13	RC SFNS IMLOIF

Table 22 – Sid $3E Negative Response Message

Response Code (hex)	Conditions
12	subFunctionNotSupported Requested sub-function is not supported
13	incorrectMessageLengthOrInvalidFormat Wrong length for the specific request

Table 23 - SId $3E Response Code

2.3.2.4 CommunicationsControl Service (SId $28)

The purpose of this service is to switch on/off the transmission and/or the reception of certain messages of (a) server(s) (e.g. application communication messages).

2.3.2.4.1 SId $28 - Request Message Defintion

The elements of the request message are described in the following paragraphs.

2.3.2.4.1.1 SId $28 - Sub-function Parameter Level Defintion

The sub-function parameter controlType contains information on how the server shall modify the communication type referenced in the communicationType parameter. See the following table.

Hex (bit 6-0)	Description	Cvt	Mnemonic
00	enableRxAndTx This value indicates that the reception and transmission of messages shall be enabled for the specified communicationType.	U	EREXTX
03	disableRxAndTx This value indicates that the reception and transmission of messages shall be disabled for the specified communicationType	U	DRXTX

Table 24 – Sid $28 sub-function parameter controlType

2.3.2.4.1.2 SId $28 - Data Parameter Defintion

The communicationType is a 1-byte value. The bits represent the communicationTypes, which can be controlled via the CommunicationControl (28 hex) service. Only the lower 2 bits (bit 0-1) of the 1-byte value are used to represent the communicationType information. The remaining bits (bit 2-7) are reserved for future definition and have to be set to zero (0). For example, a communicationType with a bit combination (Bits 1-0) of "11b" is valid and disables BOTH "normalCommunicationMessages" and "networkManagementCommunicationMessages" messages. The bit coding is shown in the following tables.

Definition
CommunicationType This parameter is used to reference the kind of communication to be controlled. The communicationType parameter is a bit-code value, which allows controlling multiple communication types at the same time.

Table 25 – Sid $28 data parameter Definition

Bit 1-0	Description	Cvt	Mnemonic
01b	normalCommunicationMessages M NCM This bit references all application-related communication (inter-application signal exchange between multiple in-vehicle servers).	U	NCM
10b	networkManagementCommunicationMessages This bit references all network management related communication.	U	NWMCM

Table 26 – Sid $28 Definition of CommunicationType byte definition

2.3.2.4.2 SId $28 - Request Message Data Bytes

See table below

A_PDU parameter	Parameter Name	Cvt	Hex Value	Mnemonic
#1	CommunicationControl Request Service ID	M	28	CC
#2	Sub-function = [control Type]	M	00, 03	LEV_ CTRLTP
#3	communicationType	M	01,02	CTP

Table 27 – Sid $28 Request Message

2.3.2.4.3 SId $28 - Positive Response Message Data Bytes

See table below.

A_PDU parameter	Parameter Name	Cvt	Hex Value	Mnemonic
#1	CommunicationControl Request resspone SId	M	68	CCPR
#2	controlType	M	00, 03	CTRLTP

Table 28 – Sid $28 Positive Response Message

2.3.2.4.4 SId $28 - Negative Response Message Data Bytes

See tables below.

A_PDU parameter	Parameter Name	Cvt	Hex Value	Mnemonic
#1	NegativeResposne SId	S	7F	NR
#2	CommunicationControl Request Service ID	M	28	CC
#3	ResponseCode = [subFunctionNotSupported incorrectMessageLengthOrInvalidFormat conditionsNotCorrect requestOutOfRange serviceNotSupportedInActiveSession]	M	 12 13 22 31 7F	 SFNS IMLOIF CRC ROOR SNSIAS

Table 29 – Sid $28 Negative Response Message

Response Code (hex)	Conditions
12	subFunctionNotSupported Requested sub-function is not supported
13	incorrectMessageLengthOrInvalidFormat Wrong length for the specific request
22	ConditionsNotCorrectOrRequestSequenceErr 1. Vehicle is moving 2. Vehicle TCS is Active
31	requestOutOfRange 1Requested communicationType value is not supported
7F	serviceNotSupportedInActiveSession The service is not enabled in the current diagnostic session.

Table 30 – Sid $28 Negative Response Codes

2.3.3 Data Transmission Functional Unit

The table below lists the services supported by the Data Transmission functional unit.

Service name	Description
ReadDataByIdentifier	The client requests to read the current value of a record identified by a provided dataIdentifier.
ReadMemoryByAddress	The client requests to read the current value of the provided memory range.
WriteDataByIdentifier	The client requests to write a record specified by a provided dataIdentifier.
Test	This is an OEM specified Service. This client sends data in its request message and the server simply take the data record from the request and return it in the response message.

Table 31 - Transmission functional unit

2.3.3.1 ReadDataByIdentifier Service (SId $22)

The ReadDataByIdentifier service allows the client to request data record values from the server identified by one or more dataIdentifiers.

The server limits the number of dataIdentifiers that can be simultaneously requested as 1.

Upon receiving a ReadDataByIdentifier request, the server will access the data elements of the records specified by the dataIdentifier parameter(s) and transmit their value in one single ReadDataByIdentifier positive response containing the associated dataRecord parameter(s).

2.3.3.1.1 SId $22 - Request Message Definition

The elements of the request message are described in the following paragraphs.

2.3.3.1.1.1 SId $22 - Sub-function Parameter Level Definition

This service does not use a sub-function parameter.

2.3.3.1.1.2 SId $22 - Data Parameter Definition

See table below. A detailed listing of supported dataIndentifiers is included in Section 4.

Data Byte	Parameter Name	Cvt	Hex Value	Mnemonic
#1	ReadDataByIdentifier Request Service Id	M	22	RDBLI
#2 #3	dataIdentifier = [byte#1 (MSB) byte#2]	M	 00-FF 00-FF	DID HB LB

Table 32 - ReadDataByIdentifier Request Message

2.3.3.1.2 SId $22 - Positive Response Message Data Bytes

See table below.

A_PDU parameter	Parameter Name	Cvt	Hex Value	Mnemonic
#1	**readDataByIdentifier Request Service Id**	**S**	**62**	**RDBI**
#2 #3	dataIdentifier =[byte#1 (MSB) byte#2]	M	 00-FF 00-FF	DID HB LB
#4 : #n	datarecord#1 = [data#1 : data#k	 M : C	 xx, : xx	DREC_ DATA_1 : DATA_k

Table 33 - ReadDataByIdentifier Positive Response Message

2.3.3.1.3 SId $22 - Negative Response Message Data Bytes

See tables below.

A_PDU parameter	Parameter Name	Cvt	Hex Value	Mnemonic
#1	**negativeResponse Service Id**	**S**	**7F**	**NR**
#2	**readDataByIdentifier Request Service Id**	**M**	**22**	**REI**
#3	ResponseCode = [incorrectMessageLengthOrInvalidFormat requestOutOfRange]	M	 13 31	 IMLOIF ROOR

Table 34 - ReadDataByIdentifier Negative Response Message

Hex	Conditions
13	incorrectMessageLengthOrInvalidFormat Wrong length for the specific request
31	RequestOutOfRange This response code will be sent if: Any DataIdentifier in the request is invalid (reserved or not defined).

Table 35 - Diagnostic Services $22 Response Code

2.3.3.2 ReadMemoryByAddress Service (SId $23)

The ReadMemoryByAddress service allows the client to request memory data from the server via provided starting address and size of memory to be read.

The readMemoryByAddress request message requests memory data from the server identified by the parameters memoryAddress and memorySize.

This diagnostic service will be used to read most of RAM, some portion of flash memory in a physically address request. DBC 1200 will deny access to any other memory address.

2.3.3.2.1 SId $23 - Request Message definition

The elements of the request message are described in the following paragraphs.

2.3.3.2.1.1 SId $23 - Sub-function Parameter Level Definition

This service does not use a sub-function parameter.

2.3.3.2.1.2 SId $23 - Data Parameter Definition

The following data-parameters are defined for this service.

Definition
addressAndLengthFormatIdentifier This parameter is a one byte value with each nibble encoded separately bit 7 - 4: number of bytes of the memorySize parameter. Support 1 byte format bit 3 - 0: number of bytes of the memoryAddress parameter. Support 3 byte format
memoryAddress The parameter memoryAddress is the starting address of server memory from which data is to be retrieved. The number of bytes used for this address is defined by the low nibble (bit 3 - 0) of the addressFormatIdentifier. Byte#3 in the memoryAddress parameter is always the least significant byte of the address being referenced in the server.
memorySize The parameter memorySize in the ReadMemoryByAddress request message specifies the number of bytes to be read starting at the address specified by memoryAddress in the server's memory. The number of bytes used for this size is defined by the high nibble (bit 7 - 4) of the addressFormatIdentifier. The maximum number of bytes can be requested is 31 bytes.

Table 36 - ReadMemoryByAddress data parameter defintion

2.3.3.2.2 SId $23 - Request Message Data Bytes

See table below.

Data Byte	Parameter Name	Cvt	Hex Value	Mnemonic
#1	readMemoryByAddress Request	M	23	RMBA
#2	addressAndLengthFormatIdentifier	M	13	ALFID
#3 #4 #5	memoryAddress[] = [byte#1 (MSB) byte#2 byte#3]	M : C1	00-FF 00-FF 00-FF	MA_ B1 B2 B3
#6	memorySize =[byte#1]	M	01-1F	MS B1

Table 37 - ReadMemoryByAddress Request Message

2.3.3.2.3 SId $23 - Positive Response Message Data Bytes

See table below.

Data Byte	Parameter Name	Cvt	Hex Value	Mnemonic
#1	readMemoryByAddress Positive Response	S	63	RMBAPR
#2 : #n	dataRecord[] = [data#1 : data#m]	M : U	00-FF : 00-FF	DREC_ DATA_1 : DATA_m

Table 38 - ReadMemoryByAddress Positive Response Message

2.3.3.2.4 SId $23 - Negative Response Message Data Bytes

See tables below.

Data Byte	Parameter Name	Cvt	Hex Value	Mnemonic
#1	negativeResponse Service Id	S	7F	NR
#2	readMemoryByAddressRequest Service Id	M	23	RMBA
#3	ResponseCode = [incorrectMessageLengthOrInvalidFormat requestOutOfRange]	M	 13 31	 IMLOIF ROOR

Table 39 - ReadMemoryByAddress Negative Response Message

Hex	Conditions
13	SubFunctionNotSupported-InvalidFormat Request message has wrong bytes count.
31	RequestOutOfRange Any memory address within the interval [$MA, ($MA + $MS -$1)] is invalid (i.e. not in RAM or readable FlashMemory) or restricted. The memorySize parameter in the request message is greater than the maximum value supported by the ECU. The specified addressAndLengthFormatIdentifier is not valid

Table 40 - Diagnostic Services $23 Response Code

2.3.3.3 WriteDataByIdentifier Service (SId $2E)

The WriteDataByIdentifier service allows the client to write information into the server at an internal location specified by the provided data identifier. The WriteDataByIdentifier service is used by the client to write a dataRecord to a server. The data is identified by a dataIdentifier, and may or may not be secured. It is the vehicle manufacturer's responsibility that the server conditions are met when performing this service.

2.3.3.3.1 SId $2E - Request Message Definition

The elements of the request message are described in the following paragraphs.

2.3.3.3.1.1 SId $2E - Sub-function Parameter Level Definition

This service does not use a sub-function parameter.

2.3.3.3.1.2 SId $2E - Data Parameter Definition

The following data-parameters are defined for this service.

Definition
dataIdentifier This parameter identifies the server data record that the client is requesting to write to
dataRecord This parameter provides the data record associated with the dataIdentifier that the client is requesting to write to.

Table 41 - Request message data parameter definition

2.3.3.3.2 SId $2E - Request Message Data Bytes

See table below.

Data Byte	Parameter Name	Cvt	Hex Value	Mnemonic
#1	WriteDataByIdentifier Request Service Id	M	2E	WDBLI
#2 #3	dataIdentifier [] =[byte#1(MSB) byte#2]	M	 00-FF 00-FF	DID_ HB LB
#4 : #n	dataRecord[] = [data#1 : data#m]	 M : M	 xx : xx	DREC_ DATA_1 : DATA_m

Table 42 - WriteDataByIdentifier Request Message

2.3.3.3.3 SId $2E - Positive Response Message Data Bytes

See table below.

Data Byte	Parameter Name	Cvt	Hex Value	Mnemonic
#1	WriteDataByIdentifier Positive Response Service Id	S	6E	WDBLIPR
#2 #3	dataIdentifier [] =[byte#1(MSB) byte#2]	M	 00-FF 00-FF	DID_ HB LB

Table 43 - WriteDataByIdentifier Positive Response Message

2.3.3.3.4 SId $2E - Negative Response Message Data Bytes

See tables below.

Data Byte	Parameter Name	Cvt	Hex Value	Mnemonic
#1	negativeResponse Service Id	S	7F	NR
#2	WriteDataByIdentifier Request Service Id	M	2E	WDBLI
#3	ResponseCode = [incorrectMessageLengthOrInvalidFormat ConditionsNotCorrect requestOutOfRange securityAccessDenied generalProgrammingFailure reqCorrectlyRcvd-RspPending serviceNotSupportedInActiveSession]	M	 13 22 31 33 72 78 7F	 IMLOIF CNC ROOR SAD GPF RCR-RP SNSIAS

Table 44 - WriteDataByIdentifier Negative Response Message

Hex	Conditions
13	incorrectMessageLengthOrInvalidFormat This response code will be sent if the length of the data in the request message does not match the size expected by the node.
22	ConditionsNotCorrect This response code shall be sent if the operating conditions of the ECU are such that it can't perform the required action (e.g., the data for a DataIdentifier is stored in EEPROM and an EEPROM failure has occurred). This response code will occur if the node (ECU) internal conditions do not allow the writing of any DataIdentifier/ IdentificationOption. The system is currently in initialization. The system is currently in shutdown process. The system is currently in SafetySystemDiagnosticSession. The system is currently in InputOutputControlByIdentifier service active stage. The system is currently busy at modifying memory. The vehicle is moving.
31	RequestOutOfRange This response code will be sent if: The DataIdentifier in the request message is not supported.
33	securityAccessDenied This negative response code is used for when security is not yet granted to perform the required service.
72	generalProgrammingFailure Request can not be serviced or system has error while try to change the EEPROM contents.
78	ReqCorrectlyRcvd-RspPending This response code shall be sent if writing the data to the address(es), which are referenced by the DataIdentifier, takes longer than $P2_{can_server}$ ms.
7F	serviceNotSupportedInActiveSession The service is not enabled in the current diagnostic session.

Table 45 - Diagnostic Services $2E Response Code

2.3.3.4 Test service (SId $B0)

This service is defined by OEM. It is only used for specific test purposes.

This service simply take the data record from the request and return it in the response message. The length of the requestRecord may be chosen to suit the test to be performed. Maximum length of the message should not exceed 30 bytes in order to not impose additional size of message buffer area.

2.3.3.4.1 SId $B0 - Request Message definition

The elements of the request message are described in the following paragraphs.

2.3.3.4.1.1 Sid $B0 - Sub-function Parameter Level definition

This service has one sub-function parameter.

Hex (bit 6-0)	Description	Mnemonic
01	Echo request The server takes recorded data from the client request and returns it in the response message.	ER

Table 46 - Request message sub function definition

2.3.3.4.1.2 SId $B0 - Data parameter definition

This service does not support data parameters in the request message.

2.3.3.4.2 SId $B0 - Request Message Data Bytes

See table below.

Data Byte	Parameter Name	Cvt	Hex Value	Mnemonic
#1	TestService request service Service Id	M	B0	TS
#2	sub-function= [Echo request]	M	01	ER
	RequestRecord = [			RR
#3	data#1	U	00~FF	DATA_1
:	:	:	:	:
#n	data#m]	U	00~FF	DATA_m

Table 47 - TestService Request Message

2.3.3.4.3 SId $B0 - Positive Response Message

See table below. Note that C depends on RequestRecord in request message. It should have the same content and length as the request message when it is short or equal to 30 bytes of data. (m <= 30)

Data Byte	Parameter Name	Cvt	Hex Value	Mnemonic
#1	TestService positive response service Id, SIDPR	S	F0	TSPR
#2	sub-function= [Echo request]	M	01	ER
	RequestRecord = [			RR
#3	data#1	C	00~FF	DATA_1
:	:	:	:	:
#n	data#m]	C	00~FF	DATA_m

Table 48 - TestService Positive Response Message

2.3.3.4.4 SId $B0 - Negative Response Message

See tables below.

Data Byte	Parameter Name	Cvt	Hex Value	Mnemonic
#1	negativeResponse Service Id	S	7F	NR
#2	TestService Request Service Id	M	B0	WDBLI
#3	ResponseCode = [	M		
	subFunctionNotSupported		12	SFNS
	incorrectMessageLengthOrInvalidFormat		13	IMLOIF
	ConditionsNotCorrect		22	CNC
	serviceNotSupportedInActiveSession		7F	SNSIAS
	]			

Table 49 - Testservice Negative Response Message

Hex	Conditions
12	subFunctionNotSupported Requested sub-function is not supported
13	incorrectMessageLengthOrInvalidFormat Request has Data longer than 30 bytes
22	ConditionsNotCorrect Request is functional addressed
7F	serviceNotSupportedInActiveSession The service is not enabled in the current diagnostic session.

Table 50 - Diagnostic Services $B0 Response Code

2.3.4 Stored Data Transmission functional unit

The table below lists the services supported by the Stored Data Transmission functional unit.

Service name	Description
ClearDiagnosticInformation	The client requests from the server to clear all or a group of the diagnostic information stored.
ReadDiagnosticTroubleCodesBy-Status	The client requests from the server the transmission of both, number of DTC and values of the diagnostic trouble codes depending on their status.

Table 51 - Stored Data Transmission functional unit

2.3.4.1 ClearDiagnosticInformation Service (SId $14)

The clearDiagnosticInformation service is used by the client (tester) to clear diagnostic information in the server's (ECU's) memory. If the parameter the server shall send a positive response when the ClearDiagnosticInformation service is completely processed. The server shall send a positive response even if no DTC's are stored. If a server supports multiple copies of DTC status information in memory (e.g. one copy in RAM and one copy in EEPROM), the server shall clear the copy used by the ReadDTCInformation status reporting service followed by the remaining copy.

This service will clear the DTC status bit(s) and failure Record Information stored in RAM and permanent memories immediately after receiving the valid ClearDiagnosticInformation request. A system reset will be triggered by the valid request. The parameters used for Fault Detection; e.g. timers, flags and variables are also reset. The amount of time taken to clear the diagnostic related information depends on the stored fault information. If the conditions responsible for fault detection are still present after this request is processed, the fault will be logged again.

2.3.4.1.1 SId $14 - Request Message Definition

The elements of the request message are described in the following paragraphs.

2.3.4.1.1.1 SId $14 - Sub-function Parameter Level Definition

This service does not use a sub-function parameter.

2.3.4.1.1.2 SId $14 - Data parameter Definition

The following data-parameters are defined for this service.

groupOfDTC
This parameter contains a 3-byte value indicating the group of DTC's (e.g., Powertrain, Body, Chassis) or the particular DTC to be cleared.

Table 52 - Request message data parameter definition

2.3.4.1.2 SId $14 - Request Message Data Bytes

See table below.

Data Byte	Parameter Name	Cvt	Hex Value	Mnemonic
#1	clearDiagnosticInformation Request Service Id		14	CDTCI
#2	groupOfDTC[] = [			GODTC_
#3	groupOfDTCHighByte		00-FF*	HB
#4	groupOfDTCMiddleByte		00-FF*	MB
	groupOfDTCLowByte		00-FF*	LB
	]			
* All group FF FF FF, chassis C1F FF FF group, any DTC in C1 group, will be possible to get a positive response. Any other, will get negative response.				

Table 53 - ClearDiagnosticInformation Request Message

2.3.4.1.3 SId $14 - Positive Response Message Data Bytes

See table below.

Data Byte	Parameter Name	Cvt	Hex Value	Mnemonic
#1	clearDiagnosticInformation Positive Response Sid	S	54	CDIPR

Table 54 - ClearDiagnosticInformation Positive Response Message

2.3.4.1.4 SId $14 - Negative Response Message Data Bytes

See tables below.

Data Byte	Parameter Name	Cvt	Hex Value	Mnemonic
#1	negativeResponse Service Id	S	7F	NR
#2	clearDiagnosticInformation Request Service Id	M	14	CDTCI
#3	ResponseCode = [	M		
	generalReject		10	GR
	incorrectMessageLengthOrInvalidFormat		13	IMLOIF
	ConditionsNotCorrect		22	CNC
	requestOutOfRange		31	ROOR
	reqCorrectlyRcvd-RspPending		78	RCR-RP
	serviceNotSupportedInActiveSession		7F	SNSIAS
	]			

Table 55 - ClearDiagnosticInformation Negative Response Message

Hex	Conditions
10	**generalReject** Request can not be serviced system has error while try to clear fault information.
13	**incorrectMessageLengthOrInvalidFormat** Request has wrong length.
22	**ConditionsNotCorrect** This response code will occur if the node (ECU) internal conditions do not allow deletion of the diagnostic information stored in the node. These are: The vehicle is moving (about 5 kph); The system is currently in initialization or shutdown process. The system is currently in SafetySystemDiagnosticSession. The system is currently in I/O Control. The system is in the process of changing some Memory content.
31	**requestOutOfRange** Wrong groupOfDTC values in request
78	**reqCorrectlyRcvd-RspPending** System is busy processing the clear code request which takes longer than $P2_{can_server}$ ms.
7F	**serviceNotSupportedInActiveSession** The service is not enabled in the current diagnostic session.

Table 56 - Diagnostic Services $14 Response Code

2.3.4.2 ReadDTCInformation Service (SId $19)

This service allows a client to read the status of server resident Diagnostic Trouble Code (DTC) information from any server, or group of servers within a vehicle. Unless otherwise stated, the server shall return DTC information. In this service we allow the client to do the following:

- Retrieve the number of DTC's matching a client defined DTC status mask (at the point of the request),
- Retrieve the list of all DTC's matching a client defined DTC status mask,
- Retrieve the status of all DTC's supported by the server,

This service uses a sub-function to determine which type of diagnostic information the client is requesting. Further details regarding each sub-function parameter are provided in the following sections.

Note:

- Diagnostic trouble codes being reported under this service are not ordered by the time they were detected.
- When there are no DTC’s stored the ECU will also return a single positive response.

2.3.4.2.1 SId $19 - Request Message Definition

The elements of the request message are described in the following paragraphs.

2.3.4.2.1.1 SId $19 - Sub-function Parameter Level Definition

The sub-function parameters are used by this service to select one of the DTC report types specified in Table 58 - Request message sub function definition. Explanations and usage of the possible levels are detailed below (suppressPosRspMsgIndicationBit (bit 7) not shown).

Hex (bit 6-0)	Description	Mnemonic
01	**reportNumberOfDTCByStatusMask** This parameter specifies that the server shall transmit to the client the number of DTC's matching a client defined status mask.	RNODTCBSM
02	**reportDTCByStatusMask** This parameter specifies that the server shall transmit to the client a list of DTC's and corresponding statuses matching a client defined status mask.	RDTCBSM

Table 57 - Request message sub function definition

2.3.4.2.1.2 Sid $19 - Data parameter Definition

The following data-parameters are defined for this service.

Definition
DTCStatusMask The DTCStatusMask contains eight (8) DTC status bits. The definitions for each of the eight (8) bits can be found in Table 59 - Definition of statusOfDTC values. This byte is used in the request message to allow a client to request DTC information for the DTC's whose status matches the DTCStatusMask. A DTC's status matches the DTCStatusMask if any one of the DTC's actual status bits is set to '1' and the corresponding status bit in the DTCStatusMask is also set to '1' (i.e., if the DTCStatusMask is bit-wise logically ANDed with the DTC's actual status and the result is non-zero, then a match has occurred). If the client specifies a status mask that contains bits that the server does not support, then the server shall process the DTC information using only the bits that it does support.

Table 58 - Request data parameter definition

2.3.4.2.2 DTC Status Bit Definitions

See table below.

Bit	Description of statusOfDTC-Response	Mnemonic	Bit support
0	testFailed	WIRS	Yes
1	testFailedThisOperationCycle	CDTCSPU	Yes
2	pendingDTC	TNPSCPU	No
3	confirmedDTC	HDTC	Yes
4	testNotCompletedSinceLastClear	TFSDTCC	Yes
5	testFailedSinceLastClear	TNPSDTCC	Yes
6	testNotCompletedThisOperationCycle	CDTC	Yes
7	warningIndicationRequested	DTCSBC	No

Table 59 - Definition of statusOfDTC values

2.3.4.2.3 SId $19 - Request Message Data Bytes

See tables below.

Data Byte	Parameter Name	Cvt	Hex Value	Mnemonic
#1	ReadDTCInformation request Service Id	M	19	RDTCI
#2	sub-function = [reportNumberOfDTCByStatusMask reportDTCByStatusMask]	M	 01 02	LEV_ RNODTCBSM RDTCBSM
#3	DTCStatusMask	M	00-FF	DTCSM

Table 60 - ReadDiagnosticTroubleCodesByStatus Request Message

2.3.4.2.4 SId $19 - Positive Response Message Data Bytes

See tables below.

Data Byte	Parameter Name	Cvt	Hex Value	Mnemonic
#1	ReadDTCInformation response Service Id	M	59	RDTCIPR
#2	reportType = [reportNumberOfDTCByStatusMask]	M	01	LEV_ RNODTCBSM
#3	DTCStatusAvailabilityMask	M	7B	DTCSAM
#4	DTCFormatIdentifier = ISO14229-1DTCFormat]	M	01	DTCFID_ 14229-1DTCF
 #5 #6	DTCCount[] = [DTCCountHighByte DTCCountLowByte]	 M M	 00-FF 00-FF	DTCC_ DTCCHB DTCCLB

Table 61 - ReadDiagnosticTroubleCodesByStatus Positive Response Message

Data Byte	Parameter Name	Cvt	Hex Value	Mnemonic
#1	ReadDTCInformation response Service Id	M	59	RDTCIPR
#2	reportType = [reportDTCByStatusMask]	M	02	LEV_ RDTCBSM
#3	DTCStatusAvailabilityMask	M	7B	DTCSAM
 #4 #5 #6 #7 #8 #9 #10 #11 : #n-3 #n-2 #n-1 #n	DTCAndStatusRecord[] = [DTCHighByte#1 DTCMiddleByte#1 DTCLowByte#1 statusOfDTC#1 DTCHighByte#2 DTCMiddleByte#2 DTCLowByte#2 statusOfDTC#2 : DTCHighByte#m DTCMiddleByte#m DTCLowByte#m statusOfDTC#m]	 C_1 C_1 C_1 C_1 C_2 C_2 C_2 C_2 : C_2 C_2 C_2 C_2	 00-FF 00-FF 00-FF 00-FF 00-FF 00-FF 00-FF 00-FF : 00-FF 00-FF 00-FF 00-FF	DTCASR_ DTCHB DTCMB DTCLB SODTC DTCHB DTCMB DTCLB SODTC : DTCHB DTCMB DTCLB SODTC
C_1: This parameter is only present if DTC information is available to be reported. C_2: This parameter is only present if more than one DTC information is available to be reported.				

Table 62 - Response message definition - sub-function = reportDTCByStatusMask, reportSupportedDTCs

2.3.4.2.5 SId $19 - Negative Response Message Data Bytes

See tables below.

Data Byte	Parameter Name	Cvt	Hex Value	Mnemonic
#1	negativeResponse Service Id	S	7F	NR
#2	readDiagnosticTroubleCodesByStatus Request SId	M	19	RDTCI
#3	ResponseCode = [subFunctionNotSupported incorrectMessageLengthOrInvalidFormat reqCorrectlyRcvd-RspPending]	M	 12 13 78	 SFNS IMLOIF RCR-RP

Table 63 - ReadDiagnosticTroubleCodesByStatus Negative Response Message

Hex	Conditions
12	**subFunctionNotSupported** Requested sub-function is not supported
13	**incorrectMessageLengthOrInvalidFormat** Wrong length for the specific request
78	**ReqCorrectlyRcvd-RspPending** This response code shall be sent if writing the data to the address(es), which are referenced by the DataIdentifier, takes longer than $P2_{can_server}$ ms.

Table 64 - Diagnostic Services $18 Response Code

2.3.5 InputOutput Control Functional Unit

The table below lists the services supported by the InputOutput functional unit.

Service name	Description
InputOutputControlByLocalIdentifier	The client requests the control of an input/output specific to the server.

Table 65 - InputOutput Control functional unit

2.3.5.1 InputOutputControlByIdentifier Service (SId $2F)

The InputOutputControlByIdentifier service is used allow external control the actuator outputs of the DBC 1200 PB EHCU.

The client request message contains a dataIdentifier to reference a group of signals of the server.

The server shall send a positive response message if the request message can be successfully executed.

Exiting Device Control

Some of the components excited by I/O Control commands will remain active for a while (settling time) even after they have been de-activated by I/O Control command. When any of the following conditions are met, the system will begin to terminate this service. The system will, however, remain in the I/O Control until it is determined that the commanded component(s) is /

are no longer active. The system will then return control to normal operation. During the component settling time, the system is allowed to go back to this service. The exiting conditions are:

1. Positively responded to a new diagnostic session request.

2. Any Output Control limit is exceeded. See section 2.3.5.1.4 Negative Response Message Data Bytes

3. The active Session is timed out per S3server.

This service is only supported if requested by Physical address mode.

2.3.5.1.1 SId $2F - Request Message Definition

The elements of the request message are described in the following paragraphs.

2.3.5.1.1.1 SId $2F - Sub-function Parameter Level Definition

This service does not use a sub-function parameter.

2.3.5.1.1.2 SId $2F - Data Parameter Definition

The following data-parameters are defined for this service.

For detailed information on the dataIdentifer byte #2 and ControlOptionRecord#1, please see "DID $3100 - $31FF Device Control" in Section 4.

Definition
dataIdentifier This parameter identifies a server local input signal(s), internal parameter(s) or output signal(s). The applicable range of values for this parameter can be found in the table of dataIdentifiers defined in Section **Error! Reference source not found. Error! Reference source not found.**. In our implementation the high byte of the dataIdentifier is used as the "Control Packet ID" or CPID and the low byte is used as the control enable bytes. However it is not used as a enable control of the particular device through this service but a control ON enable byte.
controlOptionRecord The controlOptionRecord of each dataIdentifier consists of one or multiple bytes (controlState#1 to controlState #m). ControlState#1 can be used as an additional controlState byte

Table 66 - Request message data parameter definition

2.3.5.1.2 SId $2F - Request Message Data Bytes

See table below.

Data Byte	Parameter Name	Cvt	Hex Value	Mnemonic
#1	inputInputOutputControlByIdentifier Request Sid	M	2F	IOCBLI
	dataIdentifier #1 [] = [	M		IOI
#2	byte #1 (MSB)		31	B1
#3	byte #2 (LSB)]		00-FF	B2
#4	ControlOptionRecord #1 = [			CSR
#5	ControlStatus#1	M	00-FF	CS
	ControlStatus#2	M	00-FF	CS
	ControlStatus#3	M	00-FF	CS
	ControlStatus#4]	M	00-FF	CS

Table 67 - InputOutputControlByIdentifier request message

2.3.5.1.3 SId $2F - Positive Response Message Data Bytes

See table below.

Data Byte	Parameter Name	Cvt	Hex Value	Mnemonic
#1	inputInputOutputControlByIdentifier Positive Response Sid	S	6F	IOCBLIPR
#2	dataIdentifier #1 [] = [	M		IOI
#3	byte #1 (MSB)		31	B1
	byte #2 (LSB)		00-FF	B2
	]			

Table 68 - InputOutputControlByIdentifier positive response message

2.3.5.1.4 Sid $2F - Negative Response Message Data Bytes

See tables below.

Data Byte	Parameter Name	Cvt	Hex Value	Mnemonic
#1	negativeResponse Service Id	S	7F	NR
#2	inputInputOutputControlByIdentifier Request Sid	M	2F	IOCBLI
#3	ResponseCode = [	M		
	incorrectMessageLengthOrInvalidFormat		13	IMLOIF
	ConditionsNotCorrect		22	CNC
	RequestOutOfRange		31	ROOR
	serviceNotSupportedInActiveSession		7F	SNSIAS
	VehicleSpeedLimitExceeded		88	VSLE
	MotorOrSolenoidThermalLimitExceeded		86	MOSTLE
	SystemVoltageOutOfRange		93	SVOOR
	]			

Table 69 - InputOutputControlByIdentifier negative response message

Hex	Condition
13	**incorrectMessageLengthOrInvalidFormat** Wrong length for the specific request
22	**ConditionsNotCorrect** This response code shall be sent if the operating conditions of the ECU are such that it can't perform the required action. The system is currently in initialization or shutdown process. The system is in the process of changing memory. .
31	**RequestOutOfRange** Wrong inputOutputLocalIdentifier
7F	**serviceNotSupportedInActiveSession** The service is not enabled in the current diagnostic session.
88	**VehicleSpeedLimitExceeded** This code will be returned any time the device exits I/O Control due to an unsafe condition. That is, the vehicle speed ≥ 5 KPH, while the device's is not currently in SafetySystemDiagnosticSession. Please note that this response can be send unsolicited.
86	**MotorOrSolenoidThermalLimitExceeded** This code will be returned any time the device exits I/O Control due to its internal protection algorithms. That is, any of the solenoids or motor has been commanded active for an excessive period of time that may begin to damage the device. Please note that this response can be send unsolicited.
93	**SystemVoltageOutOfRange** This code will be returned any time the device exits I/O Control due to the ignition voltage not been sufficient Please note that this response can be send unsolicited.

Table 70 - Diagnostic Services $2F Response Code

2.3.6 Upload Download Functional Unit

The table below lists the services supported by the Upload Download Functional Unit. These services can be utilized to program application software and calibration files into the DBC1200 PB ECU. Note that on vehicle programming can only be accomplished if all nodes on the vehicle comply with the Communications Control Services.

Service	Description
RequestDownload	The client requests the negotiation of a data transfer from the client to the server.
TransferData	The client transmits data to the server (download) or requests data from the server (upload).

Table 71 - Upload Download functional unit

2.3.6.1 RequestDownload Service (SId $34)

The requestDownload service is used by the client to initiate a data transfer from the client to the server (download).

After the server has received the requestDownload request message the server shall take all necessary actions to receive data before it sends a positive response message.

This service is only supported if requested by Physical address mode. . The ECU must be unlocked before a request for this service is accepted (see Mode $27 Request SecurityAccess).

2.3.6.1.1 SId $34 - Request Messages Defintion

The elements of the request message are described in the following paragraphs.

2.3.6.1.1.1 SId $34 - Sub-function Paramter Level Definition

This service does not use a sub-function paramter

2.3.6.1.1.2 SId $34- Data Paramter Defintion

See table below.

Definition
dataFormatIdentifier Information not used.This data parameter is a one byte value with each nibble encoded separately. The high nibble specifies the "compressionMethod, and the low nibble specifies the "encryptingMethod". The value 00 hex specifies that no compressionMethod nor encryptingMethod is used. Values other than 00 hex are vehicle manufacturer specific.
addressAndLengthFormatIdentifier This parameter is a one byte value with each nibble encoded separately bit 7 - 4: number of bytes of the memorySize parameter. bit 3 - 0: number of bytes of the memoryAddress parameter. This parameter is a one byte value with each nibble encoded bit 7 - 4: Length (number of bytes) of the memorySize parameter bit 3 - 0: Length (number of bytes) of the memoryAddress parameter
memoryAddress The parameter memoryAddress is the starting address of the server memory where the data is to be written to. The number of bytes used for this address is defined by the low nibble (bit 3 - 0) of the addressFormatIdentifier. Byte#m in the memoryAddress arameter is always the least significant byte of the address being referenced in the server. The most significant byte of the address can be used as a memoryIdentifier. An example of the use of a memoryIdentifier would be a dual processor server with 16 bit addressing and memory address overlap (when a given address is valid for either processor but yields a different physical memory device or internal and external flash is used). In this case, an otherwise unused byte within the memoryAddress parameter can be specified as a memoryIdentifier used to select the desired memory device. Usage of this functionality shall be as defined by vehicle manufacturer / system supplier. An example of the use of a memoryIdentifier would be a dual processor server with 16 bit addressing and memory address overlap (when a given address is valid for either processor but yields a different physical memory device or internal and external flash is used). In this case, an otherwise unused byte within the memoryAddress parameter can be specified as a memoryIdentifier used to select the desired memory device. Usage of this functionality shall be as defined by vehicle manufacturer / system supplier.
memorySize (unCompressedMemorySize) This parameter shall be used by the server to compare the uncompressed memory size with the total amount of data transferred during the TransferData service. This increases the programming security. The number of bytes used for this size is defined by the high nibble (bit 7 - 4) of the addressFormatIdentifier.

Table 72 - Request message data parameter definition

2.3.6.1.2 SId $34 - Request Message Data Bytes

See table below.

A_PDU parameter	Parameter Name	Cvt	Hex Value	Mnemonic
#1	RequestDownload Request Service Id	M	34	RD
#2	dataFormatIdentifier	M	00-FF	DFI_
#3	addressAndLengthFormatIdentifier	M	00-FF	ALFID
#4 : #(m-1)+4	memoryAddress[] = [byte#1 (MSB) : byte#m]	 M : C1	 00-FF 00-FF 00-FF	MA_ B1 : Bm
#n-(k-1) : #n	memorySize[] = [byte#1 (MSB) : byte#k]	 M : C2	 00-FF 00-FF 00-FF	MS_ B1 : Bk
C1: The presence of this parameter depends on address length information parameter of the addressAndLengthFormatIdentifier C2: The presence of this parameter depends on the memory size length information of the addressAndLengthFormatIdentifier.				

Table 73 - Request message data bytes

2.3.6.1.3 SId $34 -Positive Response Message

Not applicable.

2.3.6.1.4 SId $34 - Negative Response Message

See table below.

Data Byte	Parameter Name	Cvt	Hex Value	Mnemonic
#1	negativeResponse Service Id	S	7F	NR
#2	requestDownload Request Service Id	M	34	RD
#3	ResponseCode = [subFunctionNotSupported-invalidFormat ConditionsNotCorrectOrRequestSequenceErr the specified address & LengthFormatIdentifier is not valid securityAccessDenied- reqCorrectlyRcvd-RspPending reqCorrectlyRcvd-RspPending serviceNotSupportedInActiveDiagnosticMode voltageTooHigh VoltageTooLow]	M	 13 22 31 33 78 80 92 93	 SFNS-IF CNCORSE ROOR SAD-SAR RCR-RPSNSIADM VTH VTL

Table 74 - Sid $34 Negative Response Message

2.3.6.2 TransferData Service (SId $36)

The TransferData service is used by the client to transfer data either from the client to the server (download) or from the server to the client (upload).

The data transfer direction is defined by the preceding RequestDownload or RequestUpload service. If the client initiated a RequestDownload the data to be downloaded is included in the parameter(s) transferRequestParameter in the TransferData request message(s). If the client initiated a RequestUpload the data to be uploaded is included in the parameter(s) transferResponseParameter in the TransferData response message(s).

The TransferData service request includes a blockSequenceCounter to allow for an improved error handling in case a TransferData service fails during a sequence of multiple TransferData requests. The blockSequenceCounter of the server shall be initialized to one (1) when receiving a RequestDownload (34 hex) request message. This means that the first TransferData (36 hex) request message following the RequestDownload (34 hex) request message starts with a blockSequenceCounter of one (1).

2.3.6.2.1 SId $36 - Request Messages Defintion

The elements of the request message are described in the following paragraphs.

2.3.6.2.1.1 SId $36 - Sub-function Paramter Level Definition

This servcie does not use a sub-function paramter

2.3.6.2.1.2 SId $36 - Data Parameter Defintion

Not applicable.

2.3.6.2.2 SId $36 - Request Message Data Bytes

See table below.

A_Data byte	Parameter Name	Cvt	Hex Value	Mnemonic
#1	TransferData Request Service Id	M	36	TD
#2	blockSequenceCounter	M	00-FF	BSC
#3 : #n	transferRequestParameterRecord[] = [transferRequestParameter#1 : transferRequestParameter#m]	U : U	00-FF : 00-FF	TRPR_ TRTP_ : TRTP_

Table 75 - Sid $36 Request message data parameter definition

2.3.6.2.3 SId $36 - Positive Response Message

Not applicable.

2.3.6.2.4 SId $36 - Negative Response Message

See table below.

Data Byte	Parameter Name	Cvt	Hex Value	Mnemonic
#1	negativeResponse Service Id	S	7F	NR
#2	transferData Request Service Id	M	36	TD
#3	ResponseCode = [requestSequenceError serviceNotSupportedInActiveSession]	M	 24 80	 RSE SNSIAS

Table 76 - Sid $36 Negative Response Message

3 Service Scan Tool Interface

The following section describes recommended functions that should be implemented in service scan tools that will be used with the DBC 1200 PM EHCU. All of these functions utilize the ISO14229 diagnostic services that are described Section 2.0

3.1 Service Scan Tool Communications with Controller

In order to start and maintain diagnostic communication with the service scan tool and the DBC 1200 PM EHCU must receive service request at a period less than 5 seconds.

3.2 Diagnostic Trouble Code Information

The service scan tool should support the following functions for obtaining diagnostic code information from the DBC 1200 PM EHCU.

3.2.1 Read Current Diagnostics Codes

The service scan tool should have the capability to read diangostic codes using ISO14229 ReadDiagnosticTroubleCodesByStatus service - Sid $19. (See Section 2.3.4.2.) Note that the response diagnostic trouble code information will utilize the trouble code numbers that are include the DBC1200 Customer Diagnostic Code Cross Reference Table.

3.2.2 Read Enhanced Diagnostic History

The DBC 1200 PM EHCU provides enhanced diagnostic history functions that will record various vehicle and brake system parameters at the time a fault code is set. The service scan tool should have the capability to read the enhanced diagnostic codes information using the readDataByIdentifier service Sid $22 (see Section 2.3.3.1) and the corresponding supplier defined DIDSs for the preivous fault occurrences.. See Section 4.3.2 for additional description of the parameters provided by this service.

3.2.3 Clear Diagnostic Codes

The service scan tool should have the capability to clear diagnostic codes using the ISO14229 ClearDiagnosticInformation service - Sid $14. (See Section 2.3.4.1)

3.3 ECU Identification

The service scan tool should have the ability to obtain the part number information that is stored within the DBC 1200 PM EHCU memory. This part number information is used to identify the specific DBC1200 hardware and software that are installed on the vehicle. This information can be obtained using the static data identifiers described in section an the readDataByIdentifier service Sid $22 (see section 2.3.3.1)

3.4 Data Display Functions

The service scan tool should support data display functions that can show near-realtime data from the DBC1200 EHCU. Such data is useful for root causing DTCs from the DBC1200 EHCU and other related vehicle systems. The scan tool should also have the capability to record, store, and playback data from the DBC1200 EHCU. Data recording functions (also called 'Snap Shot" functions) enable the scan tool user to record and view general stauts data, and input / ouput signals when investigating customer complaints or diagnostic trouble codes Such functions can also be used to confirm the effectivensss of a repair after work has been completed.

The data display functions can be implemented in a scan tool using the ReadDataByIdentifier service - SId $22 and the supplier defined dynamic identifiers described in section 4.3.2. This data is available under all vehicle normal vehicle operating conditions.

3.5 Input / Output Device Control

Input / Output Device Control (referred to as "Device Control") enables the service technician to manually control the outputs of the DBC 1200 PM EHCU. Within the ISO14229 diagnostic service protocol, these functions are performed using InputOutputControlByLocalIdentifier service SId $2F and the corresponding data indentifiers described in section 4.3.3. The device control functions are used for normal vehicle assembly and service operations on the vehicle's brake system. Device control can also be used to investigate and root cause vehicle service issues that are caused by either the DBC1200 EHCU or other vehicle systems.

Implementation of the device control features within a service scan tool can greatly improve the quality and ease of service operations. The InputOutputControlByLocalIdentifier service for the DBC1200 PB EHCU is implemented using the following message format.

It is important to note that when a technician uses device control functions on the DBC1200 EHCU, normal ABS functions will not be available since some device control commands will restrict brake fluid flow to the the wheel brakes. Refer DID Section 4.3.3 for control status bytes details.

3.5.1 Hydraulic Output Control

3.5.1.1 Solenoid Control

Device control can be used to manually energize the solenoid valves in the DBC1200 PB EHCU. To energize any of the DBC1200 valves, the Sid $2F message must include the following data:

Bit 0 and 2 in DID byte #2 should be enabled and bit 3 is optional.

For any of the solenoids, individual bits in control state byte should be set to 1/0 for Solenoid ON/OFF control.

3.5.1.2 Pump Control

Device control can be used to manually turn-on the EBCM pump. To actuate any of the DBC1200 pump, the Sid$2F message must include the following data:

Bit 0 and 1 in DID byte #2 should be enabled.

Bit 1 Control state byte #2 can be set to 1/0 for Motor ON/Off control.

3.5.1.3 Brake System Bleed

The solenoid and pump control features can be utilized to create a brake system bleed routine. Refer to the DBC1200 Diagnostic and Service Manual for a detailed description of the bleed procedure.

The table below show a series of Sid$2F message requests that cycle the left front and right rear valves while running the pump (Important: This is an example of the Sid $2F commands – it is NOT an acceptable brake bleed procedure).

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EHCU Control Action	SId $2F Message Data Bytes (hex)									
	0	1	2	3	4	5	6	7	8	9
Energize LF & RR Release and Activate Pump	2F	31	07	82	03	00	00	00	00	00
De-energize LF & RR Release (Pump remains On)	2F	31	07	00	03	00	00	00	00	00
Energize LF Apply (Pump remains On)	2F	31	07	01	03	00	00	00	00	00
De-energize LF Apply (Pump remains On)	2F	31	07	00	03	00	00	00	00	00
Energize RR Apply (Pump remains On)	2F	31	07	40	03	00	00	00	00	00
De-energize RR Apply (Pump remains On)	2F	31	07	00	03	00	00	00	00	00
End Control (Solnoids and Pump Off)	2F	31	00	00	00	00	00	00	00	00

Table 77 - Sid $2F EHCU Control Example

4 Parameter Definitions

4.1 Parameter Coding

The parameters shall be coded with data types defined ASCII character set according to SAE2178-2.

4.2 Bit and Byte Order

Bit order: According to ISO/DIS 11898:1993/Amd.1

Byte order: Multiple byte parameters shall always be sent with the most significant byte first. For example, If a variable is two-byte long, at Data Byte #k, place the most significant byte, at Data Byte #k+1, place the least significant byte.

4.3 DataIdentifier Definition

The parameter Data identifier is used by the ReadDataByIdentifier service (SId $22) to request static or dynamic data from the server, by WriteDataByIdentifier service (SId $2E) to write data into the server or by InputOutputControlByIdentifier (Sid $2F) to device control the outputs. The $FXXX range specified by ISO is used by OEM to define the vehicle common dataIdentifier. The $30XX is OEM designated range for ABS to define the static or dynamic dataIdentifier. The $31XX is OEM designated range for ABS to use for device control service. DIDs for a specific program are a subset from the following table. Read and write access to a DID is also program dependent and customer defined. The list is not all inclusive of DIDs that are supported in the product.

If a variable is two-byte long, then place the MSB at the lower Data Byte count. For example, at Data Byte #k, place variable_a MSB, at Data Byte #k+1, place variable_a LSB.

See table below. X – indicates Read or Read/Write capable.

X – indicates Read or Read/Write capable.

Optional dataIdentifier	Definition	Read	Write
0100	OEMSpecific_1	X	
F182	applicationDataIdentificationDataIdentifier	X	X
F187	vehicleManufacturerSparePartNumberDataIdentifier	X	X
F18A	systemSupplierIdentifierDataIdentifier	X	
F18C	ECUSerialNumberDataIdentifier	X	
F190	vehicleIdentificationNumber	X	X
F192	systemSupplierECUHardwareNumber	X	
F193	systemSupplierECUHardwareVersionNumber	X	
F194	systemSupplierECUSoftwareNumber	X	
F195	systemSupplierECUSoftwareVersionNumber	X	
F199	DateECUReprogrammed	X	X
F19B	CalibrationDateInformation	X	X
F19D	ECUInstallationDateDataIdentifier	X	X
F1A0	OEMSpecific_2	X	X
F1A1	OEMSpecific_3	X	X
3001	SupplierECUTraceability	X	
3002	SupplierECUEndModelPartNumber	X	X
3003	SupplierEHCUTraceability	X	X
3004	SystemSupplierECUCalibrationNumber	X	
3005	BootModulePartNumber	X	
3006	Re-Flash Part Number and Suffix	X	X
3007	System Speed and Status Information	X	
3008	Power and Solenoid Information	X	
3009	Current And History ECU MCU DTC Information	X	
3010	Master Cylinder Pressure Sensor Information	X	
3011	VSE Sensor Enhanced Fault Information	X	
3012	System Lateral Acceleration Information	X	
3013	System Yaw Rate Information	X	
3014	System Steer Information	X	
3019	System Long Accel Information	X	
301A	EBCM Inertial Sensor Information	X	
301B	External CAN Inertial Sensor Information	X	
300A	Last Fault Diagnostic Information	X	
300B	Second Last Fault Diagnostic Information	X	
300C	Third Last Fault Diagnostic Information	X	
300D	CurrentDiagnosticDTCInformation	X	
300E	HistoryDiagnosticDTCInformation	X	
3020	ManufacturersEnableCounter	X	X
3034	systemsupplierECUCalibrationVersionNumber	X	
3100 to 311F	DeviceControl	n/a	n/a
Reserved	3015, 3016, 3017, 3018, 301C, 3021, 3022, 3023, 3024, 3025, 3026, 3027,FD11, FD12, FD13, FD14, FD15, FDFF	n/a	n/a

Table 78 – Data Indentifier Definition

4.3.1 Static DataIdentifier

The data values and format for above table items are explained in the following sections.

4.3.1.1 DID $0100 OEMSpecific_1 (Read/Write)

DID Byte Index	Definition	Display
#1	OEM Specific_1 – 1st char	ASCII
#2	OEM Specific_1 – 2nd char	ASCII
#3	OEM Specific_1 – 3rd char	ASCII
#4	OEM Specific_1 – 4th char	ASCII
#5	OEM Specific_1 – 5th char	ASCII
#6	OEM Specific_1 - 6th char	ASCII
#7	OEM Specific_1 – 7th char	ASCII
#8	OEM Specific_1 – 8th char	ASCII
#9	OEM Specific_1 – 9th char	ASCII
#10	OEM Specific_1 - 10th char	ASCII
#11	OEM Specific_1 - 11th char	ASCII
#12	OEM Specific_1 - 12th char	ASCII
#13	OEM Specific_1 – 13th char	ASCII
#14	OEM Specific_1 – 14th char	ASCII
#15	OEM Specific_1 – 15th char	ASCII
#16	OEM Specific_1 – 16th char	ASCII

Table 79 - DID $0100 OEMSpecific_1

4.3.1.2 DID $F182 applicationDataIdentificationDataIdentifier (Read/Write)

DID Byte Index	Definition	Display
#1	Application Data Identification – 1st char	ASCII
#2	Application Data Identification – 2nd char	ASCII
#3	Application Data Identification – 3rd char	ASCII
#4	Application Data Identification – 4th char	ASCII
#5	Application Data Identification – 5th char	ASCII
#6	Application Data Identification - 6th char	ASCII
#7	Application Data Identification – 7th char	ASCII
#8	Application Data Identification – 8th char	ASCII
#9	Application Data Identification – 9th char	ASCII
#10	Application Data Identification - 10th char	ASCII
#11	Application Data Identification - 11th char	ASCII
#12	Application Data Identification - 12th char	ASCII
#13	Application Data Identification – 13th char	ASCII
#14	Application Data Identification – 14th char	ASCII
#15	Application Data Identification – 15th char	ASCII
#16	Application Data Identification – 16th char	ASCII

Table 80 - DID $F182 applicationDataIdentificationDataIdentifier

4.3.1.3 DID $F187 vehicleManufacturerSparePartNumber (n byte Format) (Read/Write)

This version of the DID is utilized by applications with a part number configuration that requires up to n byte = 16 bytes. Length is variable based upon the part number length. The interpretation of the byte values (** HEX, USN, or ASCII) will depend on the customer part number format.

Note: ** Value will be HEX, USN, or ASCII as required by application.

DID Byte Index	Definition	Display
#1	SparePartNumber MSB	**
#...	SparePartNumber	**
#n-1	SparePartNumber 2nd LSB	**
#n	SparePartNumber LSB	**

Table 81 - DID $F187 vehicleManufacturerSparePartNumber

4.3.1.4 DID $F18A systemSupplierIdentifier (Read)

DID Byte Index	Definition	Display
#1	systemSupplierIdentifier MSB	ASCII
#2	systemSupplierIdentifier 2nd MSB	ASCII
#3	systemSupplierIdentifier	ASCII
#4	systemSupplierIdentifier	ASCII
#5	systemSupplierIdentifier	ASCII
#6	systemSupplierIdentifier	ASCII
#7	systemSupplierIdentifier	ASCII
#8	systemSupplierIdentifier	ASCII
#9	systemSupplierIdentifier 2nd LSB	ASCII
#10	systemSupplierIdentifier LSB	ASCII

Table 82 - DID $F18A systemSupplierIdentifier

4.3.1.5 DID $F190 vehicleIdentificationNumber (Read/Write)

DID Byte Index	Definition	Display
#1	VIN – 1st char	ASCII
#2	VIN – 2nd char	ASCII
#3	VIN – 3rd char	ASCII
#4	VIN – 4th char	ASCII
#5	VIN – 5th char	ASCII
#6	VIN - 6th char	ASCII
#7	VIN – 7th char	ASCII
#8	VIN – 8th char	ASCII
#9	VIN – 9th char	ASCII
#10	VIN - 10th char	ASCII
#11	VIN - 11th char	ASCII
#12	VIN - 12th char	ASCII
#13	VIN – 13th char	ASCII
#14	VIN – 14th char	ASCII
#15	VIN – 15th char	ASCII
#16	VIN – 16th char	ASCII
#17	VIN – 17th char	ASCII

Table 83 - DID $F190 vehicleIdentificationNumber

4.3.1.6 DID $F192 systemSupplierECUHardwareNumber (Read)

DID Byte Index	Definition	Display
#1	EBCM ECU Base Model Part Number MSB	USN
#2	EBCM ECU Base Model Part Number 2nd MSB	USN
#3	EBCM ECU Base Model Part Number 2nd LSB	USN
#4	EBCM ECU Base Model Part Number LSB	USN

Table 84 - DID $F192 systemSupplierIECUHardwareNumber

4.3.1.7 DID $F193 systemSupplierECUHardwareVersionNumber (Read)

DID Byte Index	Definition	Display
#1	EBCM ECU Base Model Part Number for Design Level Suffix MSB	ASCII
#2	EBCM ECU Base Model Part Number for Design Level Suffix LSB	ASCII

Table 85 - DID $F193 systemSupplierIECUHardwareVersionNumber

4.3.1.8 DID $F194 systemSupplierECUSoftwareNumber (Read)

DID Byte Index	Definition	Display
#1	EBCM ECU SoftwareNumber MSB	USN
#2	EBCM ECU SoftwareNumber 2nd MSB	USN
#3	EBCM ECU SoftwareNumber 2nd LSB	USN
#4	EBCM ECU SoftwareNumber LSB	USN

Table 86 - DID $F194 systemSupplierECUSoftwareNumber

4.3.1.9 DID $F195 systemSupplierECUSoftwareVersionNumber (Read)

DID Byte Index	Definition	Display
#1	EBCM ECU SoftwareNumber for Design Level Suffix MSB	ASCII
#2	EBCM ECU SoftwareNumber for Design Level Suffix LSB	ASCII

Table 87 - DID $F195 systemSupplierECUSoftwareVersionNumber

4.3.1.10 DID $F199 Date ECU Reprogrammed (Read/ Write*)

DID Byte Index	Definition	Display
#1	Year (MSB)	USN
#2	Year (LSB)	USN
#3	Month	USN
#4	Day	USN

Table 88 - DID $F199 dateECUReprogrammed

4.3.1.11 DID $F19B CalibrationDateInformation (Read/Write)

DID Byte Index	Definition	Display
#1	Calibration Date – 1st char	ASCII
#2	Calibration Date – 2nd char	ASCII
#3	Calibration Date – 3rd char	ASCII
#4	Calibration Date – 4th char	ASCII
#5	Calibration Date – 5th char	ASCII
#6	Calibration Date - 6th char	ASCII
#7	Calibration Date – 7th char	ASCII
#8	Calibration Date – 8th char	ASCII
#9	Calibration Date – 9th char	ASCII
#10	Calibration Date - 10th char	ASCII
#11	Calibration Date - 11th char	ASCII
#12	Calibration Date - 12th char	ASCII
#13	Calibration Date – 13th char	ASCII
#14	Calibration Date – 14th char	ASCII
#15	Calibration Date – 15th char	ASCII
#16	Calibration Date – 16th char	ASCII

Table 89 - DID $F19B CalibrationDateInformation

4.3.1.12 DID $F19D ECUInstallationDateDataIdentifier (Read/ Write)

DID Byte Index	Definition	Display
#1	ECU Installation Date – 1st char	ASCII
#2	ECU Installation Date – 2nd char	ASCII
#3	ECU Installation Date – 3rd char	ASCII
#4	ECU Installation Date – 4th char	ASCII

Table 90 - DID $F19D ECUInstallationDateDataIdentifier

4.3.1.13 DID $F1A0 OEMSpecific_2(Read/Write)

DID Byte Index	Definition	Display
#1	OEMSpecific_2 – 1st char	ASCII
#2	OEMSpecific_2 – 2nd char	ASCII
#3	OEMSpecific_2 – 3rd char	ASCII
#4	OEMSpecific_2 – 4th char	ASCII
#5	OEMSpecific_2 – 5th char	ASCII
#6	OEMSpecific_2 - 6th char	ASCII
#7	OEMSpecific_2 – 7th char	ASCII
#8	OEMSpecific_2 – 8th char	ASCII
#9	OEMSpecific_2 – 9th char	ASCII
#10	OEMSpecific_2 - 10th char	ASCII
#11	OEMSpecific_2 - 11th char	ASCII
#12	OEMSpecific_2 - 12th char	ASCII
#13	OEMSpecific_2 – 13th char	ASCII
#14	OEMSpecific_2 – 14th char	ASCII
#15	OEMSpecific_2 – 15th char	ASCII
#16	OEMSpecific_2 – 16th char	ASCII

Table 91 - DID $F1A0 OEMSpecific_2

4.3.1.14 DID $F1A1 OEMSpecific_3(Read/Write)

DID Byte Index	Definition	Display
#1	OEMSpecific_3 – 1st char	ASCII
#2	OEMSpecific_3 – 2nd char	ASCII
#3	OEMSpecific_3 – 3rd char	ASCII
#4	OEMSpecific_3 – 4th char	ASCII
#5	OEMSpecific_3 – 5th char	ASCII
#6	OEMSpecific_3 - 6th char	ASCII
#7	OEMSpecific_3 – 7th char	ASCII
#8	OEMSpecific_3 – 8th char	ASCII
#9	OEMSpecific_3 – 9th char	ASCII
#10	OEMSpecific_3 - 10th char	ASCII
#11	OEMSpecific_3 - 11th char	ASCII
#12	OEMSpecific_3 - 12th char	ASCII
#13	OEMSpecific_3 – 13th char	ASCII
#14	OEMSpecific_3 – 14th char	ASCII
#15	OEMSpecific_3 – 15th char	ASCII
#16	OEMSpecific_3 – 16th char	ASCII

Table 92 - DID $F1A2 OEMSpecific_3

4.3.1.15 DID $3003 SupplierEHCUTraceability (Read/Write)

DID Byte Index	Definition	Display
#1	EBCM EHCU Manufacture Code MSB	ASCII
#2	EBCM EHCU Manufacture Code MB	ASCII
#3	EBCM EHCU Manufacture Code LSB	ASCII
#4	EBCM EHCU Date Code MSB	ASCII
#5	EBCM EHCU Date Code MB	ASCII
#6	EBCM EHCU Date Code LSB	ASCII
#7	EBCM EHCU Year Code	ASCII
#8	EBCM EHCU Broadcast Code MSB	ASCII
#9	EBCM EHCU Broadcast Code 2nd MSB	ASCII
#10	EBCM EHCU Broadcast Code 2nd LSB	ASCII
#11	EBCM EHCU Broadcast Code LSB	ASCII

Table 93 - DID $3003 SupplierEHCUTraceability

4.3.1.16 DID $3020 ManufacturersEnableCounter (Read/Write)

Note: This DID is writable only if the Manufacturing Enable Counter is not equal to 0.

DID Byte Index	Definition	Display
#1	Manufacturing Enable Counter	USN

Table 94 - DID $3020 ManufacturersEnableCounter

4.3.1.17 DID $3001 SupplierECUTraceability (Read)

DID Byte Index	Definition	Display
#1	DBCM ECU Component Identifier MSB	ASCII
#2	DBCM ECU Component Identifier LSB	ASCII
#3	DBCM Manufacturing Site	ASCII
#4	DBCM ECU Supplier Code	ASCII
#5	DBCM ECU Manufacturing Code	ASCII
#6	DBCM ECU Manufacturing Date Code MSB	ASCII
#7	DBCM ECU Manufacturing Date Code 2nd MSB	ASCII
#8	DBCM ECU Manufacturing Date Code 2nd LSB	ASCII
#9	DBCM ECU Manufacturing Date Code LSB	ASCII

Table 95 - DID $3001 SupplierECUTraceability

4.3.1.18 DID $F18C ECUSerialNumberDataIdentifier

DID Byte Index	Definition	Display
#1	DBCM ECU Manufacturing Serial Number MSB	ASCII
#2	DBCM ECU Manufacturing Serial Number 2nd MSB	ASCII
#3	DBCM ECU Manufacturing Serial Number 2nd LSB	ASCII
#4	DBCM ECU Manufacturing Serial Number LSB	ASCII

Table 95 - DID $F18C ECU Serial Number Data Identifier

4.3.1.19 DID $3002 SupplierECUEndModelPartNumber

DID Byte Index	Definition	Display
#1	DBCM ECU End Model Part Number MSB	USN
#2	DBCM ECU End Model Part Number 2nd MSB	USN
#3	DBCM ECU End Model Part Number 2nd LSB	USN
#4	DBCM ECU End Model Part Number LSB	USN
#5	DBCM ECU End Model Part Number Suffix 1MSB	ASCII
#6	DBCM ECU End Model Part Number Suffix 2 LSB	ASCII

Table 96 - DID $3002 SupplierECUEndModelPartNumber

4.3.1.20 DID $3004 SystemSupplierECUCalibrationNumber

DID Byte Index	Definition	Display
#1	DBCM ECU SoftwareNumber MSB	USN
#2	DBCM ECU SoftwareNumber 2nd MSB	USN
#3	DBCM ECU SoftwareNumber 2nd LSB	USN
#4	DBCM ECU SoftwareNumber LSB	USN

Table 97 - DID $3004 SystemSupplierECUCalibrationNumber (Read)

4.3.1.21 DID $3005 BootModulePartNumber (Read)

DID Byte Index	Definition	Display
#1	Boot Module Part Number MSB	USN
#2	Boot Module Part Number 2nd MSB	USN
#3	Boot Module Part Number 2nd LSB	USN
#4	Boot Module Part Number LSB	USN
#5	Boot Module Part Number Suffix MSB	ASCII
#6	Boot Module Part Number Suffix LSB	ASCII

Table 98 - DID $3005 BootModulePartNumber (Read)

4.3.1.22 DID $3006 ReFlash Part Number and Suffix (Read/Write)

DID Byte Index	Definition	Display
#1	DBCM Re-Flash Part Number MSB	USN
#2	DBCM Re-Flash Part Number 2nd MSB	USN
#3	DBCM Re-Flash Part Number ber 2nd LSB	USN
#4	DBCM Re-Flash Part Number LSB	USN
#5	DBCM Re-Flash Part Number Suffix MSB	ASCII
#6	DBCM Re-Flash Part Number Suffix LSB	ASCII

Table 99 - DID $3006 Re-Flash Part Number and Suffix (Read/Write)

4.3.1.23 DID $3034 SystemSupplierECUCalibrationVersionNumber (Read)

DID Byte Index	Definition	Display
#1	DBCM ECU SoftwareVersionNumber MSB	ASCII
#2	DBCM ECU SoftwareVersionNumber LSB	ASCII

Table 100 - DID $3034 SystemSupplierECUCalibrationVersionNumber (Read)

4.3.2 Supplier Defined Dynamic DataIdentifier

The following DIDs are defined for the BWI controlled brake system.

4.3.2.1 DID $3007 System Speed and Status Information

DID Byte Index	Bit #	Definition	Units	Range	Reporting Resolution
1 - 2		LF Wheel Speed - unfiltered	KPH	0-255	0.0078125
3 - 4		RF Wheel Speed – unfiltered	KPH	0-255	0.0078125
5 - 6		LR Wheel speed - unfiltered	KPH	0-255	0.0078125
7 - 8		RR Wheel Speed – unfiltered	KPH	0-255	0.0078125
9 - 10		Filtered Vehicle Speed	KPH	0-255	0.015625
11		Rough Road Parameter	G	0-1	0.00392
12	7	LF State of Wheel Speed Electrical Status.	State	N/A	0 =NORMAL 1= FAILED
	6	RF State of Wheel Speed Electrical Status.	State	N/A	0 =NORMAL 1= FAILED
	5	LR State of Wheel Speed Electrical Status.	State	N/A	0 =NORMAL 1= FAILED
	4	RR State of Wheel Speed Electrical Status.	State	N/A	0 =NORMAL 1= FAILED
	3	Unused / Reserved			
	2	AWD Switch State	State	N/A	1 = AWD Active
	1	ABS Active	State	N/A	1 = True
	0	ABS Brake Switch (based upon brake sensor only)	State	N/A	1 = True
13 -14		System Voltage	VOLTS	0-19	1/1024
15 -16		Battery Voltage	VOLTS	0-19	1/1024
17		Unused	N/A	N/A	N/A
18		Unused	N/A	N/A	N/A
19		System State Operation	N/A	0-12	num

Table 101 - DID $3007 System Speed and Status Information

4.3.2.2 DID $3008 Power, Motor and Solenoid Information

DID Byte Index	Bit #	Definition	Units	Range	Reporting Resolution
1 - 2	All	Pump Motor Current	Amps	0 to 128	1/64
3 - 4	All	Unused / Reserved	N/A	N/A	N/A
5 – 6	All	Battery Voltage	Volts	0-19	1/1024
7 – 8	All	Pump Voltage Across Feedback	Volts	0-19	1/1024
9 - 10	All	System Voltage	Volts	0-19	1/1024
11 - 12	All	Motor Drive Temperature	Degrees	-40 to 215	1
13	7	Right Rear Release ABS Solenoid Activated	N/A	N/A	0=OFF 1=ON
	6	Right Rear Hold ABS Solenoid Activated	N/A	N/A	0=OFF 1=ON
	5	Left Rear Release ABS Solenoid Activated	N/A	N/A	0=OFF 1=ON
	4	Left Rear Hold ABS Solenoid Activated	N/A	N/A	0=OFF 1=ON
	3	Right Front Release ABS Solenoid Activated	N/A	N/A	0=OFF 1=ON
	2	Right Front Hold ABS Solenoid Activated	N/A	N/A	0=OFF 1=ON
	1	Left Front Release ABS Solenoid Activated	N/A	N/A	0=OFF 1=ON
	0	Left Front Hold ABS Solenoid Activated	N/A	N/A	0=OFF 1=ON
14	7-2	Unused / Reserved	N/A	N/A	N/A
	1	Pump Enable Command	N/A	N/A	0=OFF 1=ON
	0	Unused / Reserved	N/A	N/A	N/A
15 - 16		Unused / Reserved	N/A	N/A	N/A
17		Unused / Reserved	N/A	N/A	N/A
18		Unused / Reserved	N/A	N/A	N/A

Table 102 - DID $3008 Power, Motor and Solenoid Information

4.3.2.3 DID $3010 Master Cylinder Pressure Sensor Information

DID Byte Index	Bit #	Description	Units	Range	Reporting Resolution
1 - 2	all	Master Cylinder Pressure Sensor 1 Input	Volts	0-5	0.000244141
3 - 4	all	Master Cylinder Pressure Sensor 1 Input – Eng	psi	-3000 to +3000	1
5 - 6	all	Master Cylinder Pressure Sensor 1 Bias	PSI	-3000 to +3000	0.125
7 - 8	all	Master Cylinder Pressure Sensor 2 Input	Volts	0-5	0.000244141
9 - 10	all	Master Cylinder Pressure Sensor Input 2 – Eng	psi	-3000 to +3000	1
11 -12	all	Master Cylinder Pressure Sensor 2 Bias	PSI	-3000 to +3000	0.125
13 - 14	all	Centered Master Cylinder Pressure	PSI	-3000 to +3000	0.125
15	7	Master Cylinder Brake Switch	N/A	N/A	1= ON, 0 = OFF
	6	MC Brake Switch Was On	N/A	N/A	1= YES, 0=NO
	5	MC Brake Switch Was Off	N/A	N/A	1= YES, 0=NO
	4 ~3	Not Used	N/A	N/A	N/A
	2	MC Heavy Brake Switch	N/A	N/A	1= ON, 0 = OFF
	1~0	Not used	N/A	N/A	N/A
16 - 17	all	MC Pressure Sensor Power Supply Output	Volts	0 – 5	0.000244141

Table 103 - DID $3010 Master Cylinder Pressure Sensor Information

4.3.2.4 DID $3014 System Steer Information

DID Byte Index	Bit #	Definition	Units	Range	Reporting Resolution
1 - 2	all	Serial Steer Angle	Deg	-720-720	0.5
3 – 4	all	Serial Steer Rate	Deg/sec	+/- 2048	1
5 - 6	all	Steering Wheel Angle Centered	Deg	-720-720	0.5
7 - 8	all	Filtered Steering Wheel Angle Rate	Deg/sec	+/- 2048	0.125
9 - 10	all	Unused / Reserved	N/A	N/A	N/A
11	7	Steering Angle is Centered	N/A	N/A	1=YES
	6	Reverse Check	N/A	N/A	1=YES
	5	Steering Sensor Bias Failed	N/A	N/A	1=YES
	4	Unused / Reserved	N/A	N/A	N/A
	3	Steer_Sensor_Failed	N/A	N/A	1=YES
	2	VSE_Failed	N/A	N/A	1=YES
	1	VSE_Info_Valid	N/A	N/A	1=YES
	0	Vehicle_VSE_Active	N/A	N/A	1=YES
12	7	Msg_Protection_fault	N/A	N/A	1=YES
	6	Unused / Reserved	N/A	N/A	N/A
	5	Serial_Steer_Sensor_Failed	N/A	N/A	1=YES
	4	Serial_Steer_Angle Validity_Fault	N/A	N/A	1=YES
	3	Msg_Rolling_count_fault	N/A	N/A	1=YES
	2	Unused / Reserved	N/A	N/A	N/A
	1	Unused / Reserved	N/A	N/A	N/A
	0	Serial Steer Angle Normal	N/A	N/A	1=YES
13	7	Estimated_Steer_Angle_Valid	N/A	N/A	1=YES
	6	Steer_Centering_Fault	N/A	N/A	1=YES
	5	Unused / Reserved	N/A	N/A	N/A
	4	Unused / Reserved	N/A	N/A	N/A
	3	Unused / Reserved	N/A	N/A	N/A
	2	Unused / Reserved	N/A	N/A	N/A
	1	Serial_Steer_Rate_Validity_Fault	N/A	N/A	1=YES
	0	Serial_Steer_Rate_Normal	N/A	N/A	1=YES
14 – 15	all	Steering Wheel Bias	Degrees	-180 to +180	0.25
16 - 17	all	Estimated Steering Wheel Angle	Degrees	+/- 720	0.25
18 - 21	all	Unused / Reserved	N/A	N/A	N/A

Table 104 - DID $3014 System Steer Information

4.3.2.5 DID $3013 System Yaw Rate Information

DID Byte Index	Bit #	Definition	Units	Range	Reporting Resolution
1 - 2	all	Vehicle Yaw Rate - Engineering Units	deg/sec	-75 - 75	0.5
3 – 4	all	Filtered Yaw Rate	deg/sec	-75 - 75	0.5
5 - 6	all	Yaw Rate Error	deg/sec	-75 - 75	0.5
7 – 8	all	Yaw Accel	deg/s/s	+/- 2048	0.0625
9	all	Yaw Rate Bias	deg/sec	-64 - 64	0.5
10 – 11	all	Estimated_Yaw_Rate	deg/sec	+75 … -75	0.5
12 – 13	all	Yaw_Rate_Centered	deg/sec	+75 … -75	0.5
14	7	Yaw_Rate_Is_Centered	N/A	N/A	1 = YES
	6	Unused / Reserved	N/A	N/A	N/A
	5	Estimated_Yaw_Rate_is Valid	N/A	N/A	1 = YES
	4	Unused	N/A	N/A	N/A
	3	Vehicle_Stopped	N/A	N/A	1 = YES
	2	VSE_Failed	N/A	N/A	1 = YES
	1	VSE_Info_Valid	N/A	N/A	1 = YES
	0	Vehicle_VSE_Active	N/A	N/A	1 = YES
15	7	Unused / Reserved	N/A	N/A	N/A
	6	Yaw_Sensor_Error	N/A	N/A	1 = YES
	5	Yaw_Accel_Fault	N/A	N/A	1 = YES
	4	Yaw_Bias_Fault	N/A	N/A	1 = YES
	3	Unused / Reserved	N/A	N/A	N/A
	2	Unused / Reserved	N/A	N/A	N/A
	1	Yaw_Sensor_Failed	N/A	N/A	1 = YES
	0	Unused / Reserved	N/A	N/A	N/A

Table 105 - DID $3013 System Yaw Information

4.3.2.6 DID $3012 System Lateral Acceleration Information

DID Byte Index	Bit #	Definition	Units	Range	Reporting Resolution
1 - 2	all	Lateral Acceleration Input Value	g's	-4 to 4	0.00012207
3 – 4	all	Lateral Acceleration Filtered	g's	-2 to 2	0.0078125
5 – 6	all	Lateral Acceleration Centered	g's	-2 to 2	0.0078125
7	all	Lateral Accelerometer Bias	g's	-0.5 to 0.5	0.0078125
8 - 9	all	Vehicle CG Lateral Acceleration	g's	-2 to 2	0.000061035
10	7	Unused	N/A	N/A	N/A
	6	Lat_Sensor_Failed	N/A	N/A	1 = YES
	5	Unused	N/A	N/A	N/A
	4	Lat_Sensor_Error	N/A	N/A	1 = YES
	3	Vehicle_Stopped	N/A	N/A	1 = YES
	2	VSE_Failed	N/A	N/A	1 = YES
	1	VSE_Info_Valid	N/A	N/A	1 = YES
	0	Vehicle_VSE_Active	N/A	N/A	1 = YES
11	7	Lat_Jerk_Fault	N/A	N/A	1 = YES
	6	Lat_Stopped_Fault	N/A	N/A	1 = YES
	5	Lat_Bias_Fault	N/A	N/A	1 = YES
	4	Unused / Reserved	N/A	N/A	N/A
	3	Unused / Reserved	N/A	N/A	N/A
	2	Sensor_Pluasiblity_Fault	N/A	N/A	1 = YES
	1	Unused / Reserved	N/A	N/A	N/A
	0	Sensor_Plausiblity_Error	N/A	N/A	1 = YES

Table 106 - DID $3012 System Lat Information

4.3.2.7 DID $3019 System Longitudinal Acceleration Information

DID Byte Index	Bit #	Definition	Units	Range	Reporting Resolution
1 - 2	all	Unused / Reserved	N/A	N/A	N/A
3 - 4	all	Longitudinal Acceleration Input Value	g	-2 to 2	0.000488281
5 - 6	all	Biased Acclerometer Signal	g	-2 to 2	0.000488281
7 - 8	all	Unfiltered_Vehicle_Speed	kph	0 to 360	0.015625
9 – 10	all	Accelerometer_Vehicle_Speed	kph	0 to 360	0.015625
11	0	Invalid_Accelerometer_Signal	N/A	N/A	1 = YES
	1	Vehicle_Spin_Up_Pending	N/A	N/A	1 = YES
	2	Vehicle_Spin_Up	N/A	N/A	1 = YES
	3	Reverse_Check	N/A	N/A	1 = YES
	4	Spin_Up_Recovery_Condition	N/A	N/A	1 = YES
	5	Unused / Reserved	N/A	N/A	N/A
	6	Unused / Reserved	N/A	N/A	N/A
	7	Unused / Reserved	N/A	N/A	N/A

Table 107 - DID $3019 System Longitudinal Acceleration Information

4.3.2.8 DID $301A EBCM Inertial Sensor Information

DID Byte Index	Bit #	Definition	Units	Range	Reporting Resolution
1 - 2	all	Yaw Rate Sensor Input (Unfiltered)	Volts	0-5	u4p12_T
3 – 4	all	ECU_X_Axis_Accel	g's	-4…4	0.00012207
5 - 6	all	ECU_Y_Axis_Accel	g's	-4…4	0.00012207
7 - 8	all	ECU_Z_Axis_Accel	g's	-4…4	0.00012207
9 - 10	all	Lateral Acceleration - Engineering Units	g's	-1.28 ... 1.28	0.000061035
11 - 12	all	Longitudinal Acceleration - Engineering Units	g's	-1.28 ... 1.28	0.000061035
13 - 14	all	Yaw Rate Sensor Input (Unfiltered)	deg/s		
15-16	all	Unused / Reserved	N/A	N/A	N/A

Table 108 - DID $301A EBCM Inertial Sensor Information

4.3.2.9 DID $301B External CAN Inertial Sensor Information

DID Byte Index	Bit #	Definition	Units	Range	Reporting Resolution
1 - 2	all	Serial_Yaw_Rate	deg/s	+/-128	0.00390625
3	0-4	Unused / Reserved	N/A	N/A	N/A
	5	Sensor Self-Test Fault	N/A	N/A	1=YES
	6	Sensor Calibration Fault	N/A	N/A	1=YES
	7	Signal Supervision Rolling Count Fault	N/A	N/A	1=YES
4	8-9	Unused / Reserved	N/A	N/A	N/A
	10	Loss of Comm with Sensor	N/A	N/A	1=YES
	11	Data Invalid Fault	N/A	N/A	1=YES
	12-14	Unused / Reserved	N/A	N/A	N/A
	15	Normal Operation	N/A	N/A	1=YES
5 - 6	all	Serial_Lat_Accel	g's	-4…4	0.00012207
7	0-4	Unused / Reserved	N/A	N/A	NA
	5	Sensor Self-Test Fault	N/A	N/A	1=YES
	6	Sensor Calibration Fault	N/A	N/A	1=YES
	7	Signal Supervision Rolling Count Fault	N/A	N/A	1=YES
8	8-9	Unused / Reserved	N/A	N/A	N/A
	10	Loss of Comm with Sensor	N/A	N/A	1=YES
	11	Data Invalid Fault	N/A	N/A	1=YES
	12-14	Unused / Reserved	N/A	N/A	N/A
	15	Normal Operation	N/A	N/A	1=YES
9 - 10	all	Serial_Long_Accel	g's	-4…4	0.00012207
11	0-4	Unused / Reserved	N/A	N/A	N/A
	5	Sensor Self-Test Fault	N/A	N/A	1=YES
	6	Sensor Calibration Fault	N/A	N/A	1=YES
	7	Signal Supervision Rolling Count Fault	N/A	N/A	1=YES
12	8-9	Unused / Reserved	N/A	N/A	N/A
	10	Loss of Comm with Sensor	N/A	N/A	1=YES
	11	Data Invalid Fault	N/A	N/A	1=YES
	12-14	Unused / Reserved	N/A	N/A	N/A
	15	Normal Operation	N/A	N/A	1=YES

Table 109 - DID $301B External CAN Inertial Sensor Information

4.3.2.10 DID $3009 Current And History ECU MCU DTC Information

DID Byte Index	Definition	Units	Range	Reporting Resolution
1	EBCM Current MCU Fault SincePowerUp 1	N/A	N/A	1=set
2	EBCM Current MCU Fault SincePowerUp 2	N/A	N/A	1=set
3	EBCM Current MCU Fault SincePowerUp 3	N/A	N/A	1=set
4	EBCM Current MCU Fault SincePowerUp 4	N/A	N/A	1=set
5	EBCM Current MCU Fault SincePowerUp 5	N/A	N/A	1=set
6	EBCM Current MCU Fault SincePowerUp 6	N/A	N/A	1=set
7~9	Unused	N/A	N/A	N/A
10	EBCM Current ECU Fault SincePowerUp 1	N/A	N/A	1=set
11	EBCM Current ECU Fault SincePowerUp2	N/A	N/A	1=set
12	EBCM Current ECU Fault SincePowerUp 3	N/A	N/A	1=set
13	EBCM Current ECU Fault SincePowerUp 4	N/A	N/A	1=set
14	EBCM Current ECU Fault SincePowerUp 5	N/A	N/A	1=set
15~16	Unused	N/A	N/A	N/A
17	EBCM History MCU Faults 1	N/A	N/A	1=set
18	EBCM History MCU Faults 2	N/A	N/A	1=set
19	EBCM History MCU Faults 3	N/A	N/A	1=set
20	EBCM History MCU Faults 4	N/A	N/A	1=set
21	EBCM History MCU Faults 5	N/A	N/A	1=set
22	EBCM History MCU Faults 6	N/A	N/A	1=set
23~25	Unused	N/A	N/A	N/A
26	EBCM History ECU Faults 1	N/A	N/A	1=set
27	EBCM History ECU Faults 2	N/A	N/A	1=set
28	EBCM History ECU Faults 3	N/A	N/A	1=set
29	EBCM History ECU Faults 4	N/A	N/A	1=set
30	EBCM History ECU Faults 5	N/A	N/A	1=set
31	Unused	N/A	N/A	N/A
32	Unused	N/A	N/A	N/A

Table 110 - DID $3009 Current And History ECU MCU DTC Information

4.3.2.11 DID $300A Last Fault Diagnostic Information

This DID will contain information for the last stored fault code along with information about the vehicle and ESC controller state when the fault code occurred. If the fault code occurred multiple times prior to automatic fault history clearing, the vehicle and ESC controller state information will correspond to the first occurrence of the fault code.

DID Byte Index	Bit #	Definition	Units	Range	Reporting Resolution
1 - 3		Last Fault Code Set. Same format as SId $19			
4	7	TCS Active at 1st occcurence of DTC.	N/A	N/A	1=TRUE
	6	DRP Active at 1st occcurence of DTC.	N/A	N/A	1=TRUE
	5	CBD Active at 1st occcurence of DTC.	N/A	N/A	1=TRUE
	4	Unused / Reserved	N/A	N/A	N/A
	3	Vehicle ABS Active at 1st occcurence of DTC.	N/A	N/A	1=TRUE
	2	Unused / Reserved	N/A	N/A	N/A
	1	Brake Switch On at 1st occcurence of DTC.	N/A	N/A	1=TRUE
	0	Engine Running at 1st occcurence of DTC.	N/A	N/A	1=TRUE
5	7	Unused / Reserved	N/A	N/A	N/A
	6	Unused / Reserved	N/A	N/A	N/A
	5	Initialization Active at 1st occcurence of DTC.	N/A	N/A	1=TRUE
	4	Rough Road Active at 1st occcurence of DTC.	N/A	N/A	1=TRUE
	3	Steer Angle Centered at 1st occcurence of DTC.	N/A	N/A	1=TRUE
	2	VSE Enabled at 1st occcurence of DTC.	N/A	N/A	1=TRUE
	1	TCS Enabled at 1st occcurence of DTC.	N/A	N/A	1=TRUE
	0	VSE Active at 1st occcurence of DTC.	N/A	N/A	1=TRUE
6		Vehicle Speed at 1st occcurence of DTC.	kph	0 - 255	1
7		System Voltage at 1st occurrence of DTC	Volt	0 - 18	1
8		Ignition Cycles Since Most Recent occurrence of DTC	count	0 - 255	1
9		Ignition Cycles Since 1st occurrence of DTC.	count	0 - 255	1
10		Number of Occurrences of DTC	count	0 - 255	1
11		LF Wheel Speed at 1st occcurence of DTC.	kph	0 - 255	1
12		RF Wheel Speed at 1st occcurence of DTC.	kph	0 - 255	1
13		LR Wheel Speed at 1st occcurence of DTC.	kph	0 - 255	1
14		RR Wheel Speed at 1st occcurence of DTC.	kph	0 - 255	1
15 - 16		Drive cycle time at 1st occurrence of DTC	seconds	0~65535	1
17		Unused / Reserved	N/A	N/A	N/A
18		Unused / Reserved	N/A	N/A	N/A
19 - 20		Steer Angle at 1st occurrence of DTC	Deg	+/- 720	0.5
21		Vehicle Yaw Rate at 1st occcurence of DTC.	Deg/s	+/- 75	1
22		Vehicle lateral acceleration at 1st occcurence of DTC.	G's	+/- 1.28	0.015625
23		Vehicle longitudinal acceleration at 1st occcurence of DTC.	G's	+/- 1.5	0.015625
24		Accelerator pedal position at 1st occcurence of DTC.	Percent	0 -100%	0.391
25		Master cylinder pressure at 1st occcurence of DTC.	Bar	0 - 207	1
26		Ambient temperature at 1st occcurence of DTC.	Deg C	+/- 127	1
27		Unused/Reserved	N/A	N/A	N/A
28		Unused/Reserved	N/A	N/A	N/A

Table 111 - DID $300A Last Fault Diagnostic Information

4.3.2.12 DID $300B Second Last Fault Diagnostic Information

This DID will contain information for the second to last stored fault code along with information about the vehicle and ESC controller state when the fault code occurred. If the fault code occurred multiple times prior to automatic fault history clearing, the vehicle and ESC controller state information will correspond to the first occurrence of the fault code

DID Byte Index	Bit #	Definition	Units	Range	Reporting Resolution
1 - 3		Second Last Fault Code Set. Same format as SId $19			
4	7	TCS Active at 1st occcurence of DTC.	N/A	N/A	1=TRUE
	6	DRP Active at 1st occcurence of DTC.	N/A	N/A	1=TRUE
	5	CBD Active at 1st occcurence of DTC.	N/A	N/A	1=TRUE
	4	Unused / Reserved	N/A	N/A	N/A
	3	Vehicle ABS Active at 1st occcurence of DTC.	N/A	N/A	1=TRUE
	2	Unused / Reserved	N/A	N/A	N/A
	1	Brake Switch On at 1st occcurence of DTC.	N/A	N/A	1=TRUE
	0	Engine Running at 1st occcurence of DTC.	N/A	N/A	1=TRUE
5	7	Unused / Reserved	N/A	N/A	N/A
	6	Unused / Reserved	N/A	N/A	N/A
	5	Initialization Active at 1st occcurence of DTC.	N/A	N/A	1=TRUE
	4	Rough Road Active at 1st occcurence of DTC.	N/A	N/A	1=TRUE
	3	Steer Angle Centered at 1st occcurence of DTC.	N/A	N/A	1=TRUE
	2	VSE Enabled at 1st occcurence of DTC.	N/A	N/A	1=TRUE
	1	TCS Enabled at 1st occcurence of DTC.	N/A	N/A	1=TRUE
	0	VSE Active at 1st occcurence of DTC.	N/A	N/A	1=TRUE
6		Vehicle Speed at 1st occcurence of DTC.	kph	0 - 255	1
7		System Voltage at 1st occurrence of DTC	Volt	0 - 18	1
8		Ignition Cycles Since Most Recent occurrence of DTC	count	0 - 255	1
9		Ignition Cycles Since 1st occurrence of DTC.	count	0 - 255	1
10		Number of Occurrences of DTC	count	0 - 255	1
11		LF Wheel Speed at 1st occcurence of DTC.	kph	0 - 255	1
12		RF Wheel Speed at 1st occcurence of DTC.	kph	0 - 255	1
13		LR Wheel Speed at 1st occcurence of DTC.	kph	0 - 255	1
14		RR Wheel Speed at 1st occcurence of DTC.	kph	0 - 255	1
15 – 16		Drive cycle time at 1st occurrence of DTC	seconds	0~65535	1
17		Unused / Reserved	N/A	N/A	N/A
18		Unused / Reserved	N/A	N/A	N/A
19 – 20		Steer Angle at 1st occurrence of DTC	Deg	+/- 720	0.5
21		Vehicle Yaw Rate at 1st occcurence of DTC.	Deg/s	+/- 75	1
22		Vehicle lateral acceleration at 1st occcurence of DTC.	G's	+/- 1.28	0.015625
23		Vehicle longitudinal acceleration at 1st occcurence of DTC.	G's	+/- 1.5	0.015625
24		Accelerator pedal position at 1st occcurence of DTC.	Percent	0 -100%	0.391
25		Master cylinder pressure at 1st occcurence of DTC.	Bar	0 - 207	1
26		Ambient temperature at 1st occcurence of DTC.	Deg C	+/- 127	1
27		Unused/Reserved	N/A	N/A	N/A
28		Unused/Reserved	N/A	N/A	N/A

Table 112 - DID $300B Second Last Fault Diagnostic Information

4.3.2.13 DID $300C Third Last Fault Diagnostic Information

This DID will contain information for the third to last stored fault code along with information about the vehicle and ESC controller state when the fault code occurred. If the fault code occurred multiple times prior to automatic fault history clearing, the vehicle and ESC controller state information will correspond to the first occurrence of the fault code

DID Byte Index	Bit #	Definition	Units	Range	Reporting Resolution
1 - 3		Third Fault Code Set. Same format as SId $19			
4	7	TCS Active at 1st occcurence of DTC.	N/A	N/A	1=TRUE
	6	DRP Active at 1st occcurence of DTC.	N/A	N/A	1=TRUE
	5	CBD Active at 1st occcurence of DTC.	N/A	N/A	1=TRUE
	4	Unused / Reserved	N/A	N/A	N/A
	3	Vehicle ABS Active at 1st occcurence of DTC.	N/A	N/A	1=TRUE
	2	Unused / Reserved	N/A	N/A	N/A
	1	Brake Switch On at 1st occcurence of DTC.	N/A	N/A	1=TRUE
	0	Engine Running at 1st occcurence of DTC.	N/A	N/A	1=TRUE
5	7	Unused / Reserved	N/A	N/A	N/A
	6	Unused / Reserved	N/A	N/A	N/A
	5	Initialization Active at 1st occcurence of DTC.	N/A	N/A	1=TRUE
	4	Rough Road Active at 1st occcurence of DTC.	N/A	N/A	1=TRUE
	3	Steer Angle Centered at 1st occcurence of DTC.	N/A	N/A	1=TRUE
	2	VSE Enabled at 1st occcurence of DTC.	N/A	N/A	1=TRUE
	1	TCS Enabled at 1st occcurence of DTC.	N/A	N/A	1=TRUE
	0	VSE Active at 1st occcurence of DTC.	N/A	N/A	1=TRUE
6		Vehicle Speed at 1st occcurence of DTC.	kph	0 - 255	1
7		System Voltage at 1st occurrence of DTC	Volt	0 - 18	1
8		Ignition Cycles Since Most Recent occurrence of DTC	count	0 - 255	1
9		Ignition Cycles Since 1st occurrence of DTC.	count	0 - 255	1
10		Number of Occurrences of DTC	count	0 - 255	1
11		LF Wheel Speed at 1st occcurence of DTC.	kph	0 - 255	1
12		RF Wheel Speed at 1st occcurence of DTC.	kph	0 - 255	1
13		LR Wheel Speed at 1st occcurence of DTC.	kph	0 - 255	1
14		RR Wheel Speed at 1st occcurence of DTC.	kph	0 - 255	1
15 – 16		Drive cycle time at 1st occurrence of DTC	seconds	0~65535	1
17		Unused / Reserved	N/A	N/A	N/A
18		Unused / Reserved	N/A	N/A	N/A
19 - 20		Steer Angle at 1st occurrence of DTC	Deg	+/- 720	0.5
21		Vehicle Yaw Rate at 1st occcurence of DTC.	Deg/s	+/- 75	1
22		Vehicle lateral acceleration at 1st occcurence of DTC.	G's	+/- 1.28	0.015625
23		Vehicle longitudinal acceleration at 1st occcurence of DTC.	G's	+/- 1.5	0.015625
24		Accelerator pedal position at 1st occcurence of DTC.	Percent	0 -100%	0.391
25		Master cylinder pressure at 1st occcurence of DTC.	Bar	0 - 207	1
26		Ambient temperature at 1st occcurence of DTC.	Deg C	+/- 127	1
27		Unused/Reserved	N/A	N/A	N/A
28		Unused/Reserved	N/A	N/A	N/A

Table 113 - DID $300C Third Last Fault Diagnostic Information

4.3.2.14 DID $300D CurrentDiagnosticDTCInformation

DID Byte Index	Definition	Units	Range	Reporting Resolution
1	EBCM TestFailedThisOperationCycle 18 to 11	N/A	N/A	1= Set, 0=Not Set
2	EBCM TestFailedThisOperationCycle 28 to 21	N/A	N/A	1= Set, 0=Not Set
3	EBCM TestFailedThisOperationCycle 38 to 31	N/A	N/A	1= Set, 0=Not Set
4	EBCM TestFailedThisOperationCycle 48 to 41	N/A	N/A	1= Set, 0=Not Set
5	EBCM TestFailedThisOperationCycle 58 to 51	N/A	N/A	1= Set, 0=Not Set
6	EBCM TestFailedThisOperationCycle 68 to 61	N/A	N/A	1= Set, 0=Not Set
7	EBCM TestFailedThisOperationCycle 78 to 71	N/A	N/A	1= Set, 0=Not Set
8	EBCM TestFailedThisOperationCycle 88 to 81	N/A	N/A	1= Set, 0=Not Set
9	EBCM TestFailedThisOperationCycle 98 to 91	N/A	N/A	1= Set, 0=Not Set
10	EBCM TestFailedThisOperationCycle 108 to 101	N/A	N/A	1= Set, 0=Not Set
11	EBCM TestFailedThisOperationCycle 118 to 111	N/A	N/A	1= Set, 0=Not Set
12	EBCM TestFailedThisOperationCycle 128 to 121	N/A	N/A	1= Set, 0=Not Set
13	EBCM TestFailedThisOperationCycle 138 to 131	N/A	N/A	1= Set, 0=Not Set
14	EBCM TestFailedThisOperationCycle 148 to 141	N/A	N/A	1= Set, 0=Not Set
15	EBCM TestFailedThisOperationCycle 158 to 151	N/A	N/A	1= Set, 0=Not Set
16	EBCM TestFailedThisOperationCycle 168 to 161	N/A	N/A	1= Set, 0=Not Set
17	EBCM TestFailedThisOperationCycle 178 to 171	N/A	N/A	1= Set, 0=Not Set
18	EBCM TestFailedThisOperationCycle 188 to 181	N/A	N/A	1= Set, 0=Not Set
19	EBCM TestFailedThisOperationCycle 198 to 191	N/A	N/A	1= Set, 0=Not Set
20	EBCM TestFailedThisOperationCycle 208 to 201	N/A	N/A	1= Set, 0=Not Set
21	EBCM TestFailedThisOperationCycle 218 to 211	N/A	N/A	1= Set, 0=Not Set
22	EBCM TestFailedThisOperationCycle 228 to 221	N/A	N/A	1=Set, 0=Not Set

Table 114 - DID $300D CurrentDiagnosticDTCInformation

4.3.2.15 DID $300E HistoryDiagnosticDTCInformation

DID Byte Index	Definition	Units	Range	Reporting Resolution
1	EBCM HistoryDTC 18 to 11	N/A	N/A	1= Set, 0=Not Set
2	EBCM HistoryDTC 28 to 21	N/A	N/A	1= Set, 0=Not Set
3	EBCM HistoryDTC 38 to 31	N/A	N/A	1= Set, 0=Not Set
4	EBCM HistoryDTC 48 to 41	N/A	N/A	1= Set, 0=Not Set
5	EBCM HistoryDTC 58 to 51	N/A	N/A	1= Set, 0=Not Set
6	EBCM HistoryDTC 68 to 61	N/A	N/A	1= Set, 0=Not Set
7	EBCM HistoryDTC 78 to 71	N/A	N/A	1= Set, 0=Not Set
8	EBCM HistoryDTC 88 to 81	N/A	N/A	1= Set, 0=Not Set
9	EBCM HistoryDTC 98 to 91	N/A	N/A	1= Set, 0=Not Set
10	EBCM HistoryDTC 108 to101	N/A	N/A	1= Set, 0=Not Set
11	EBCM HistoryDTC 118 to 111	N/A	N/A	1= Set, 0=Not Set
12	EBCM HistoryDTC 128 to121	N/A	N/A	1= Set, 0=Not Set
13	EBCM HistoryDTC 138 to 131	N/A	N/A	1= Set, 0=Not Set
14	EBCM HistoryDTC 148 to 141	N/A	N/A	1= Set, 0=Not Set
15	EBCM HistoryDTC 158 to 151	N/A	N/A	1= Set, 0=Not Set
16	EBCM HistoryDTC 168 to 161	N/A	N/A	1= Set, 0=Not Set
17	EBCM HistoryDTC 178 to 171	N/A	N/A	1= Set, 0=Not Set
18	EBCM HistoryDTC 188 to 181	N/A	N/A	1= Set, 0=Not Set
19	EBCM HistoryDTC 198 to 191	N/A	N/A	1= Set, 0=Not Set
20	EBCM HistoryDTC 208 to 201	N/A	N/A	1= Set, 0=Not Set
21	EBCM HistoryDTC 218 to 211	N/A	N/A	1= Set, 0=Not Set
22	EBCM HistoryDTC 228 to 221	N/A	N/A	1= Set, 0=Not Set

Table 115 - DID $300E HistoryDiagnosticDTCInformation

4.3.2.16 DID $3011 VSE Sensor Enhanced Fault Information

DID Byte Index	Bit #	Definition	Units	Range	Reporting Resolution
1	all	Last DTC that caused VSE disabled	N/A	N/A	BCD
2	all	Lateral Accel Value at Last VSE disable DTC	g	-1.28 ~1.28	0.015625
3	all	Yaw Rate Value at Last VSE disable DTC	deg/sec	-75 to +75	1
4 - 5	all	Steer Sensor Value at Last VSE disable DTC	Degrees	-720 ~ 720	0.25
6	all	Unused / Reserved			
7	all	Yaw Rate Bias at Last VSE disable DTC	deg/sec	-20 to +20	0.5
8	all	Lat Error Value at Last VSE disable DTC	g	-1.2 to +1.2	0.015625
9	all	Lat Accel Sensor Bias at Last VSE disable DTC	g	-0.5 to +0.5	0.0078125
10	all	Steer Sensor Bias at Last VSE disable DTC	degrees	-128 to +127	1
11 - 12	all	ECI I-sensor Status at Last VSE disable DTC	N/A	N/A	N/A
	15~11	Not used	N/A	N/A	1 = YES
	10	snapshot_data_available	N/A	N/A	1 = YES
	9	status_SPI_fault	N/A	N/A	1 = YES
	8	temp_SPI_fault	N/A	N/A	1 = YES
	7	accel_z_SPI_fault	N/A	N/A	1 = YES
	6	accel_y_SPI_fault	N/A	N/A	1 = YES
	5	accel_x_SPI_fault	N/A	N/A	1 = YES
	4	rate_SPI_fault	N/A	N/A	1 = YES
	3	illegal_sw_status	N/A	N/A	1 = YES
	2	startup_fault	N/A	N/A	1 = YES
	1	internal_sensor_fault	N/A	N/A	1 = YES
	0	normal	N/A	N/A	1 = YES

Table 116 - DID $3011 VSE Sensor Enhanced Fault Information

4.3.3 DID $3100 ~ $311F DeviceControl

4.3.3.1 DataIdentifier #1 Definition

DID Byte Index	Bit #	Definition	Units	Range	Command Resolution
1	all	$31			
2	7~5	Not Used.	N/A	N/A	N/A
	4	Enable Sensor Power Supply Control Command	N/A	N/A	Enable = 1 Disable = 0
	3	Enable Solenoid PWM Control Command	N/A	N/A	Enable = 1 Disable = 0
	2	Enable solenoid ON-Off Control Command	N/A	N/A	Enable = 1 Disable = 0
	1	Enable Pump Motor Control Command	N/A	N/A	Enable = 1 Disable = 0
	0	Enable system Relay Control Command	N/A	N/A	Enable = 1 Disable = 0

Table 117 - dataIdentifier #1 byte #2 Definition

4.3.3.2 Control Option Record bytes definition

DID Byte Index	Bit #	Definition	Units	Range	Command Resolution
1	7	Right Rear Release Solenoid On-Off Control Command	N/A	N/A	Enable = 1 Disable = 0
	6	Right Rear Apply Solenoid On-Off Control Command	N/A	N/A	Enable = 1 Disable = 0
	5	Left Rear Release Solenoid On-Off Control Command	N/A	N/A	Enable = 1 Disable = 0
	4	Left Rear Apply Solenoid On-Off Control Command	N/A	N/A	Enable = 1 Disable = 0
	3	Right Front Release Solenoid On-Off Control Command	N/A	N/A	Enable = 1 Disable = 0
	2	Right Front Apply Solenoid On-Off Control Command	N/A	N/A	Enable = 1 Disable = 0
	1	Left Front Release Solenoid On-Off Control Command	N/A	N/A	Enable = 1 Disable = 0
	0	Left Front Apply Solenoid On-Off Control Command	N/A	N/A	Enable = 1 Disable = 0
2	7	Primary Prime Solenoid On-Off Control Command	N/A	N/A	Enable = 1 Disable = 0
	6	Secondary Prime Solenoid On-Off Control Command	N/A	N/A	Enable = 1 Disable = 0
	5	Primary Isolation Solenoid On-Off Control Command	N/A	N/A	Enable = 1 Disable = 0
	4	Secondary Isolation Solenoid On-Off Control Command	N/A	N/A	Enable = 1 Disable = 0
	3	Not Used.	N/A	N/A	
	2	Sensor Power Supply	N/A	N/A	Enable = 1 Disable = 0
	1	First Pump Motor On/Off Control Command	N/A	N/A	Enable = 1 Disable = 0
	0[1]	System Relay Control Command	N/A	N/A	Enable = 1 Disable = 0
3 – 4 [2]	All	Pump Motor Voltage Command Please note the max of pump motor voltage is limited by the system voltage.	Volt	0 - 32	0.00098

[1] For Byte #2 bit 0 value

[2] For the system without the motor pump voltage value control, the value has no effect to the system. **#3 and 4 bytes:** parameter is optional, if missing from the request message, fill the bytes with $FFFF as if received the two bytes with data $FFFF. In other words, the length either be 5 bytes or 7 bytes is accepted.

Table 118 - Control Option Record #1

4.3.4 Supplier Reserved DIDs

The following SId $2E and SId$ 22 DIDs are reserved for BWI engineering development and manufacturing use. They are only functional for the EBCM physical address ID.

$3015

$3016

$3017

$3018

$301C

$3021

$3022

$3023

$3024

$3025

$3026

$3027

FD11

FD12

FD13

FD14

FD15

FDFF

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